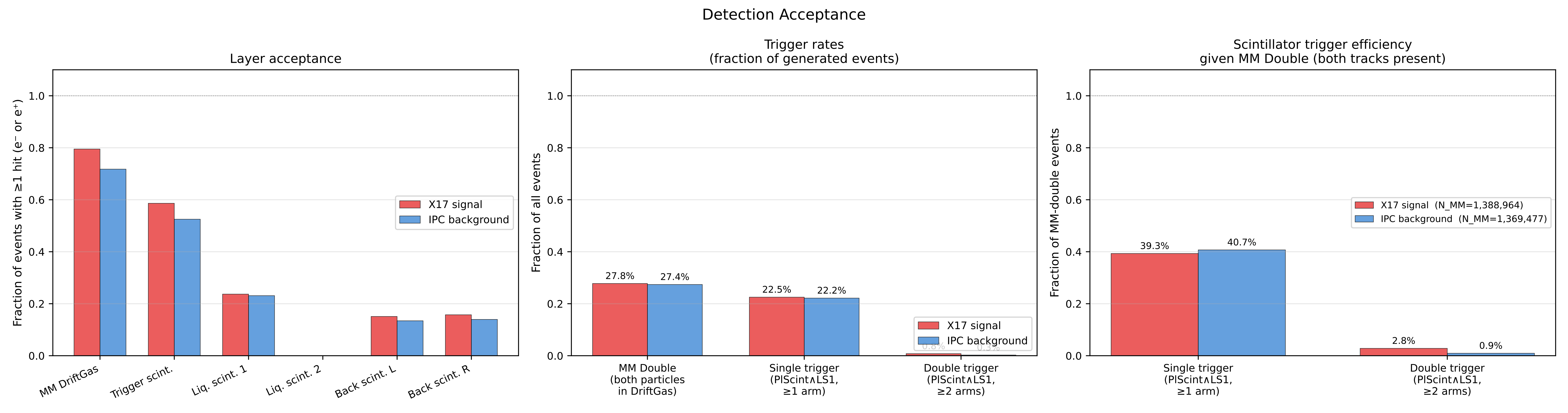
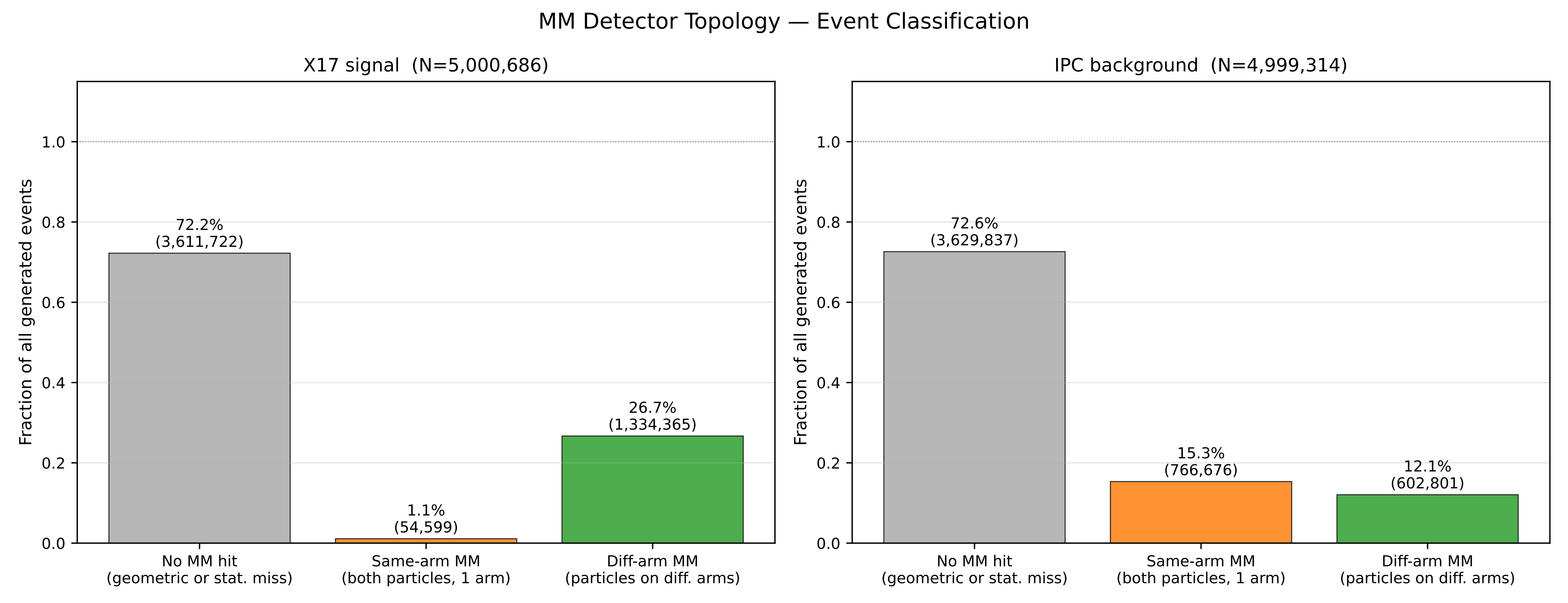
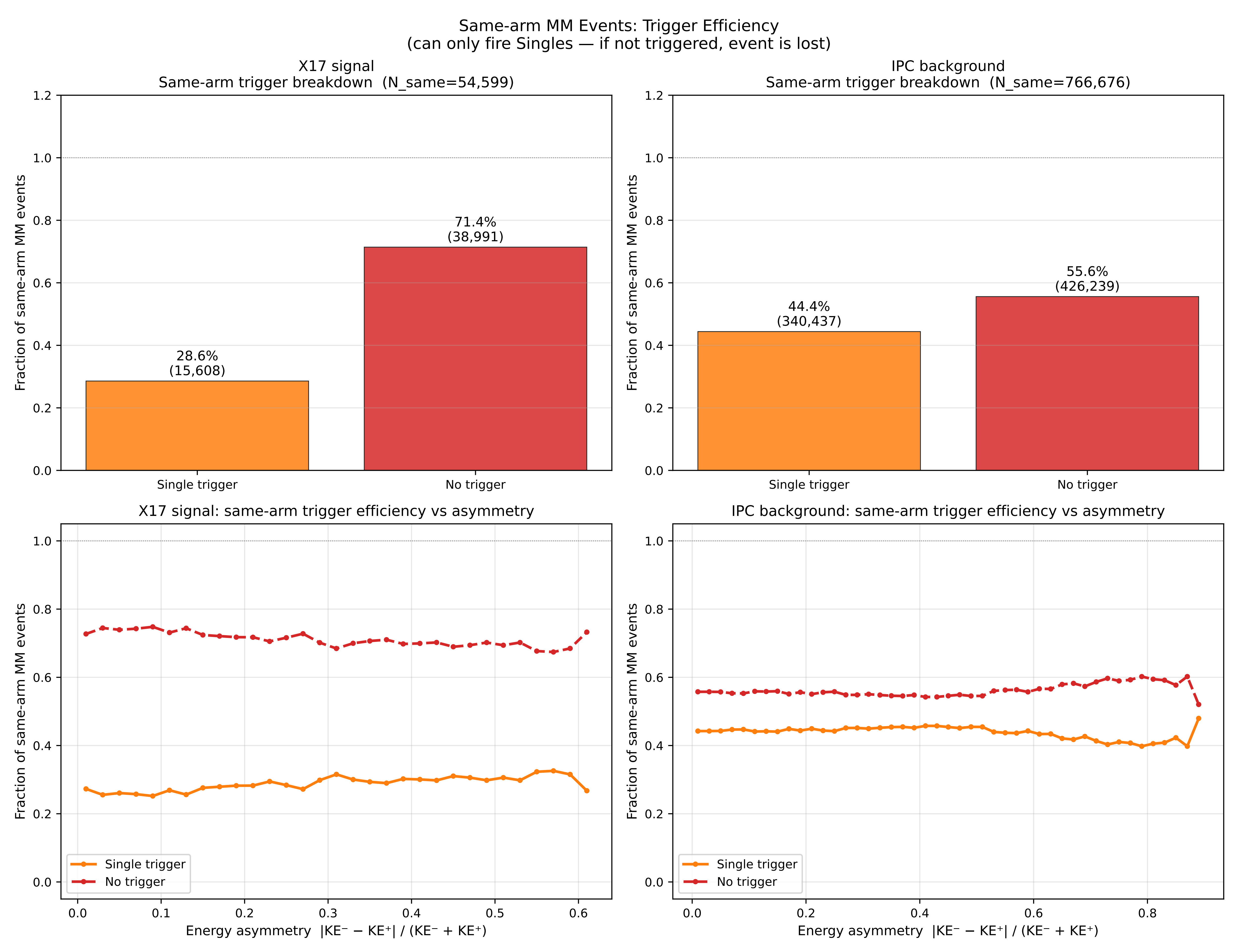
```
MX17 Pair Simulation Analysis

────────────────────────────────────────────────────
X17 signal  (5,000,686 events)
  MM Double (both tracks)  :  27.8%
  LiqScint_1 reached       :  23.7%
  LiqScint_2 reached       :  0.0%
  Single trigger (all evt) :  22.5%
  Double trigger (all evt) :  0.8%
  Single | MM Double       :  39.3%
  Double | MM Double       :  2.8%

────────────────────────────────────────────────────
IPC background  (4,999,314 events)
  MM Double (both tracks)  :  27.4%
  LiqScint_1 reached       :  23.1%
  LiqScint_2 reached       :  0.0%
  Single trigger (all evt) :  22.2%
  Double trigger (all evt) :  0.3%
  Single | MM Double       :  40.7%
  Double | MM Double       :  0.9%
```

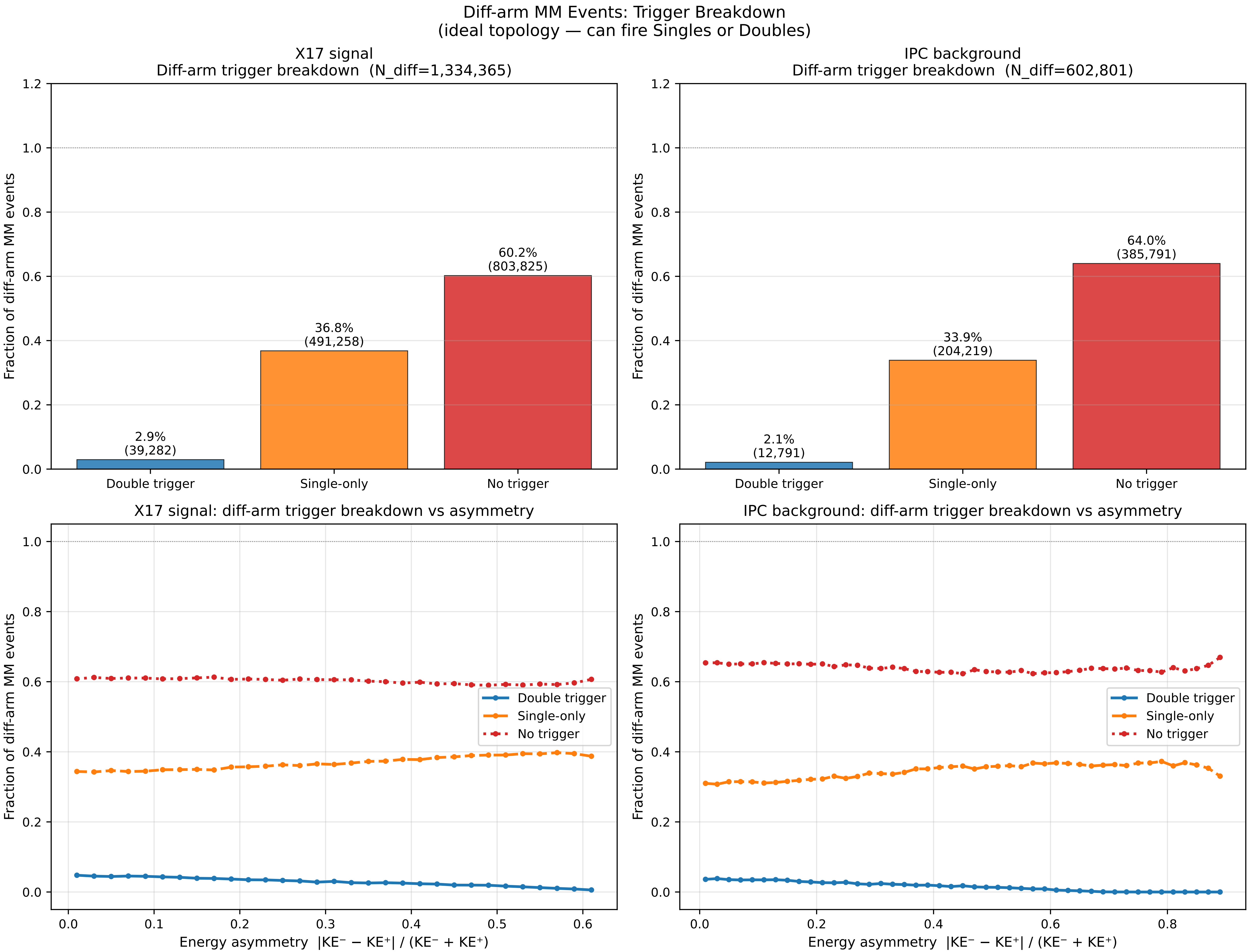


Detection Acceptance
Layer acceptance
Fraction of events with ≥1 hit (e⁻ or e⁺)
MM DriftGas
Trigger scint.
Liq. scint. 1
Liq. scint. 2
Back scint. L
Back scint. R
X17 signal
IPC background
Trigger rates
(fraction of generated events)
Fraction of all events
27.8%
27.4%
22.5%
22.2%
0.8%
0.3%
MM Double
(both particles
in DriftGas)
Single trigger
(PlScint∧LS1,
≥1 arm)
Double trigger
(PlScint∧LS1,
≥2 arms)
Scintillator trigger efficiency
given MM Double (both tracks present)
Fraction of MM-double events
X17 signal (N_MM=1,388,964)
IPC background (N_MM=1,369,477)
39.3%
40.7%
2.8%
0.9%

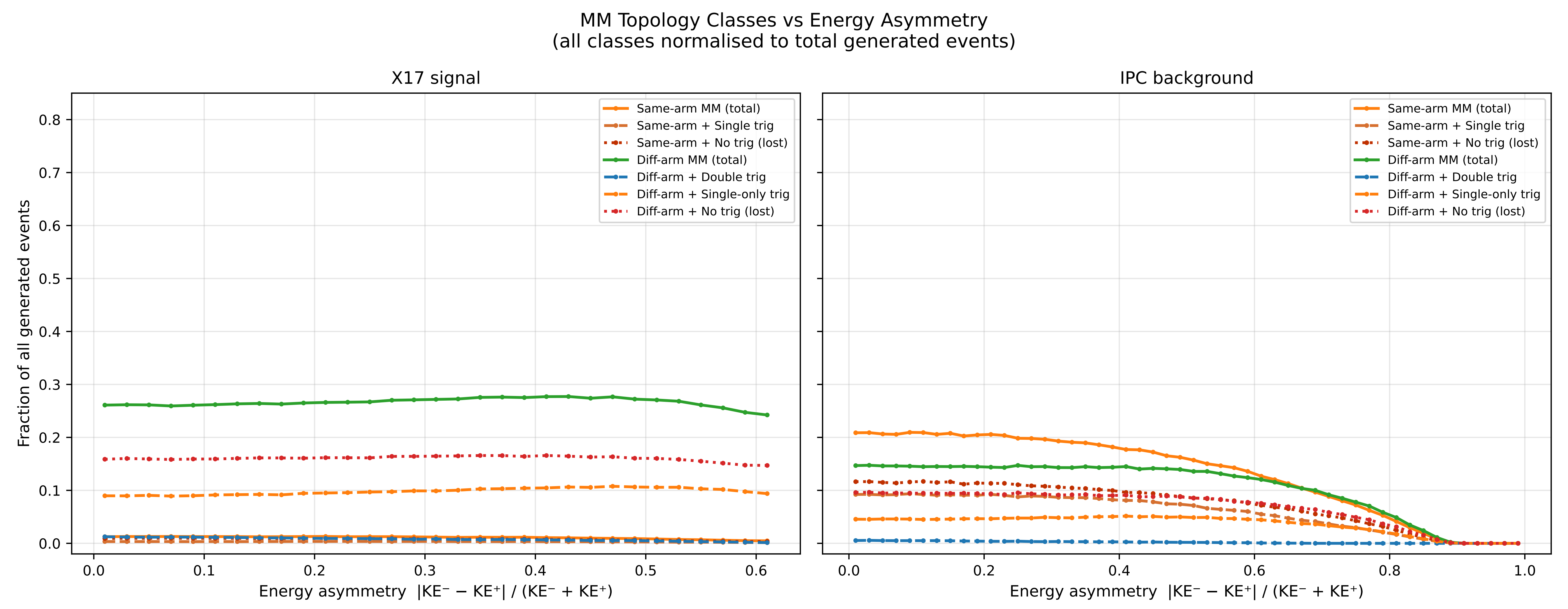

MM Detector Topology — Event Classification
X17 signal (N=5,000,686)
IPC background (N=4,999,314)
Fraction of all generated events
1.0
0.8
0.6
0.4
0.2
0.0
72.2%
(3,611,722)
1.1%
(54,599)
26.7%
(1,334,365)
72.6%
(3,629,837)
15.3%
(766,676)
12.1%
(602,801)
No MM hit
(geometric or stat. miss)
Same-arm MM
(both particles, 1 arm)
Diff-arm MM
(particles on diff. arms)

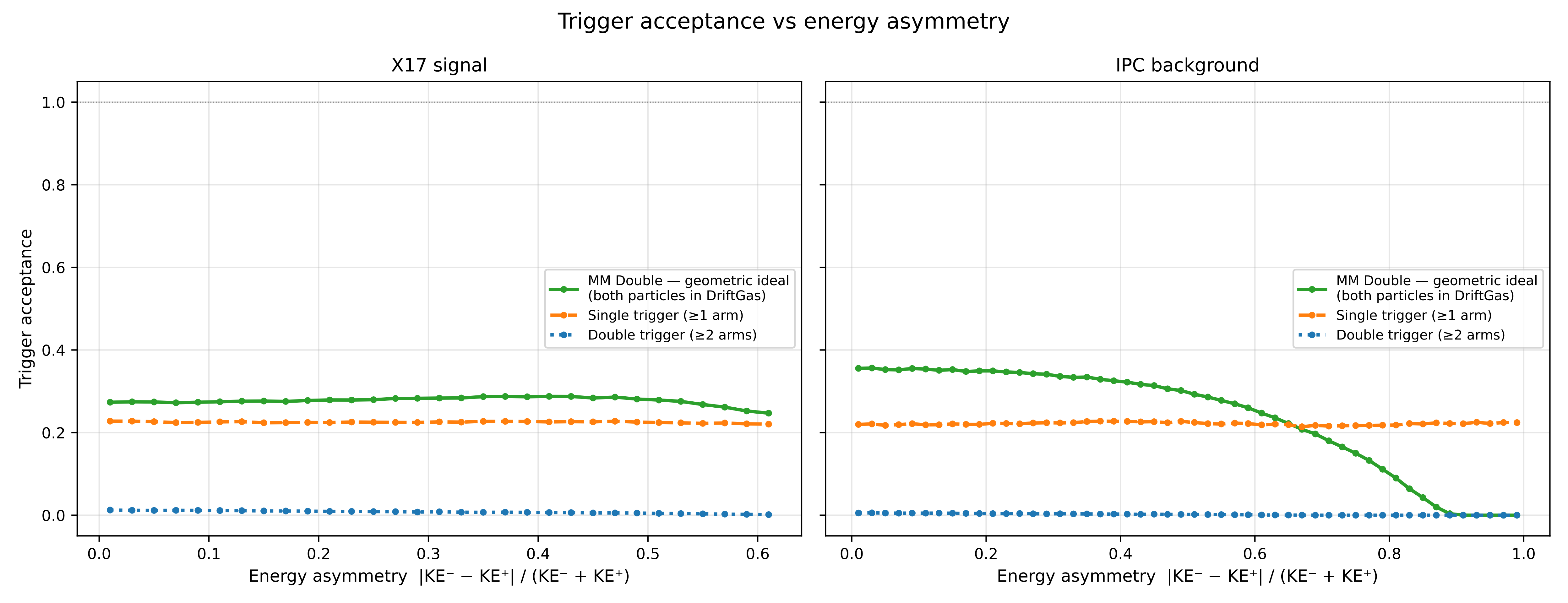


Same-arm MM Events: Trigger Efficiency
(can only fire Singles — if not triggered, event is lost)
X17 signal
Same-arm trigger breakdown (N_same=54,599)
28.6%
(15,608)
71.4%
(38,991)
Single trigger
No trigger
Fraction of same-arm MM events
IPC background
Same-arm trigger breakdown (N_same=766,676)
44.4%
(340,437)
55.6%
(426,239)
Single trigger
No trigger
Fraction of same-arm MM events
X17 signal: same-arm trigger efficiency vs asymmetry
Single trigger
No trigger
Fraction of same-arm MM events
Energy asymmetry |KE⁻ − KE⁺| / (KE⁻ + KE⁺)
IPC background: same-arm trigger efficiency vs asymmetry
Single trigger
No trigger
Fraction of same-arm MM events
Energy asymmetry |KE⁻ − KE⁺| / (KE⁻ + KE⁺)

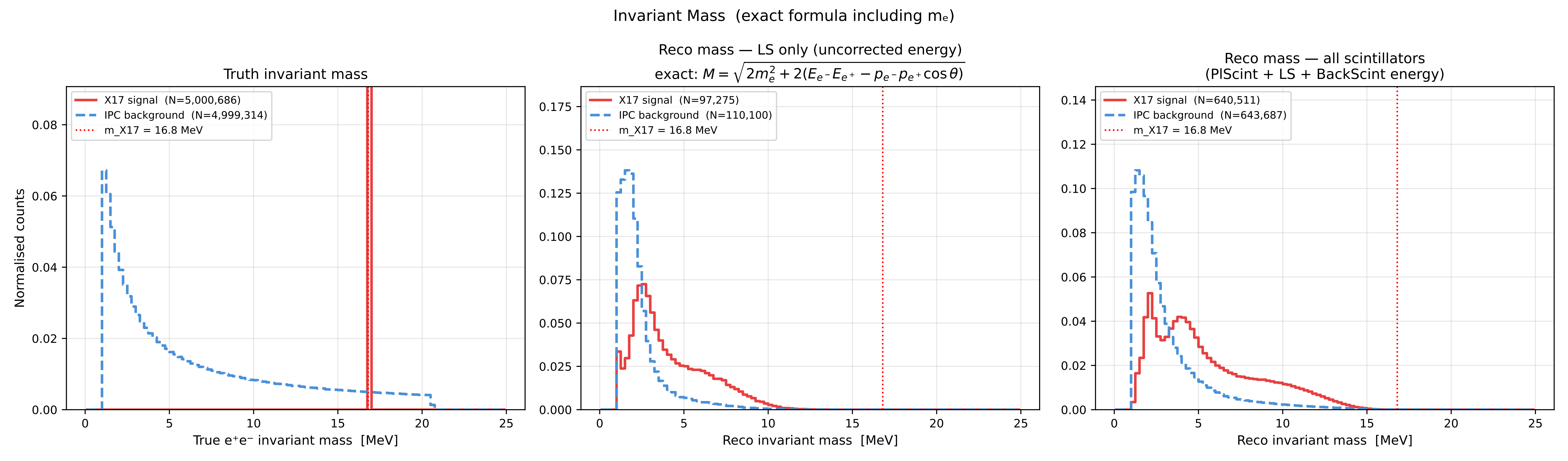


Diff-arm MM Events: Trigger Breakdown
(ideal topology — can fire Singles or Doubles)
X17 signal
Diff-arm trigger breakdown (N_diff=1,334,365)
Fraction of diff-arm MM events
2.9%
(39,282)
36.8%
(491,258)
60.2%
(803,825)
Double trigger
Single-only
No trigger
IPC background
Diff-arm trigger breakdown (N_diff=602,801)
Fraction of diff-arm MM events
2.1%
(12,791)
33.9%
(204,219)
64.0%
(385,791)
Double trigger
Single-only
No trigger
X17 signal: diff-arm trigger breakdown vs asymmetry
Fraction of diff-arm MM events
Double trigger
Single-only
No trigger
Energy asymmetry |KE⁻ − KE⁺| / (KE⁻ + KE⁺)
IPC background: diff-arm trigger breakdown vs asymmetry
Fraction of diff-arm MM events
Double trigger
Single-only
No trigger
Energy asymmetry |KE⁻ − KE⁺| / (KE⁻ + KE⁺)

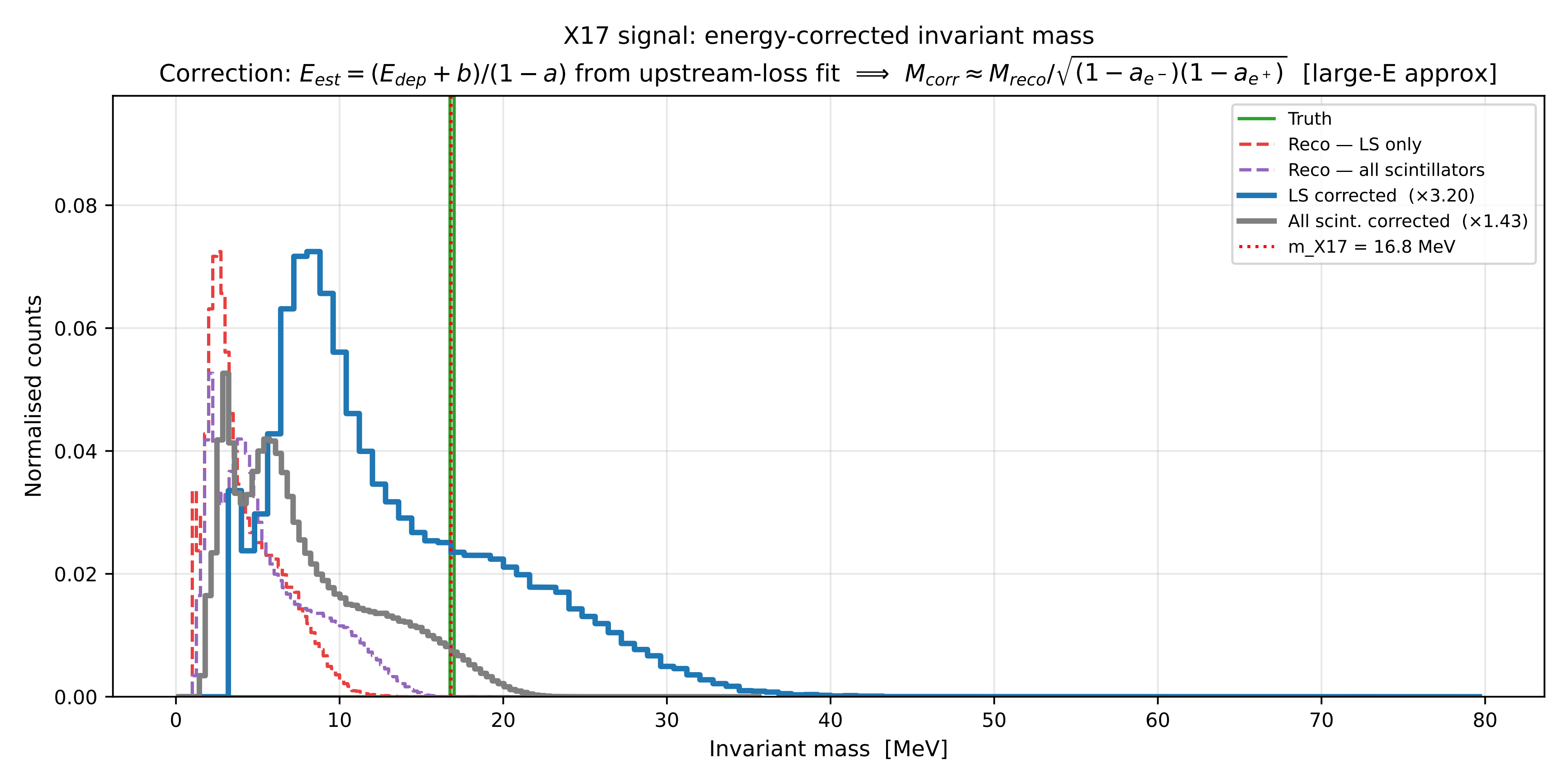

MM Topology Classes vs Energy Asymmetry
(all classes normalised to total generated events)
X17 signal
IPC background
Same-arm MM (total)
Same-arm + Single trig
Same-arm + No trig (lost)
Diff-arm MM (total)
Diff-arm + Double trig
Diff-arm + Single-only trig
Diff-arm + No trig (lost)
Fraction of all generated events
Energy asymmetry |KE⁻ − KE⁺| / (KE⁻ + KE⁺)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
1.0



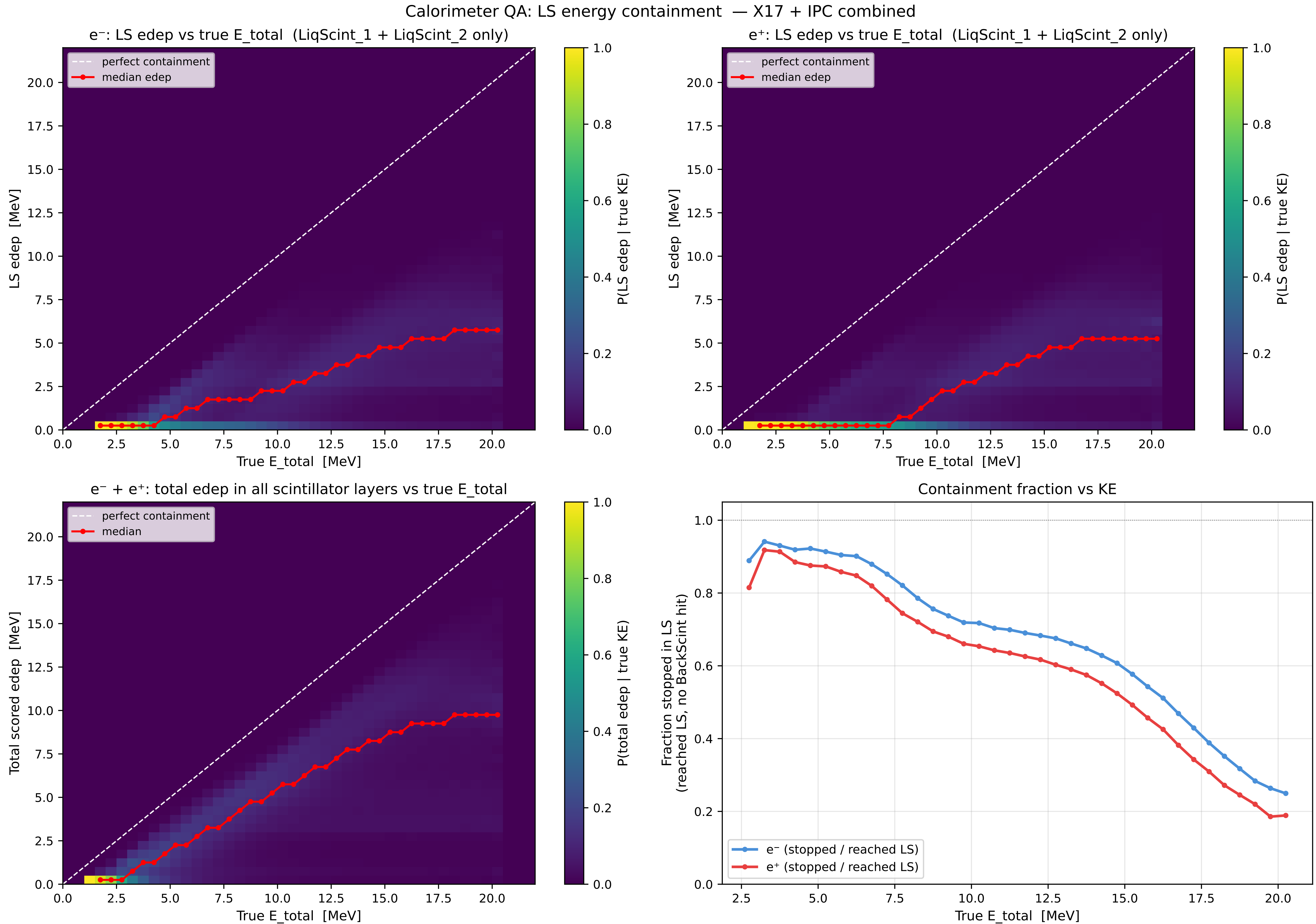
Trigger acceptance vs energy asymmetry
X17 signal
IPC background
Trigger acceptance
Energy asymmetry |KE⁻ − KE⁺| / (KE⁻ + KE⁺)
MM Double — geometric ideal
(both particles in DriftGas)
Single trigger (≥1 arm)
Double trigger (≥2 arms)
0.0
0.2
0.4
0.6
0.8
1.0
0.1
0.3
0.5

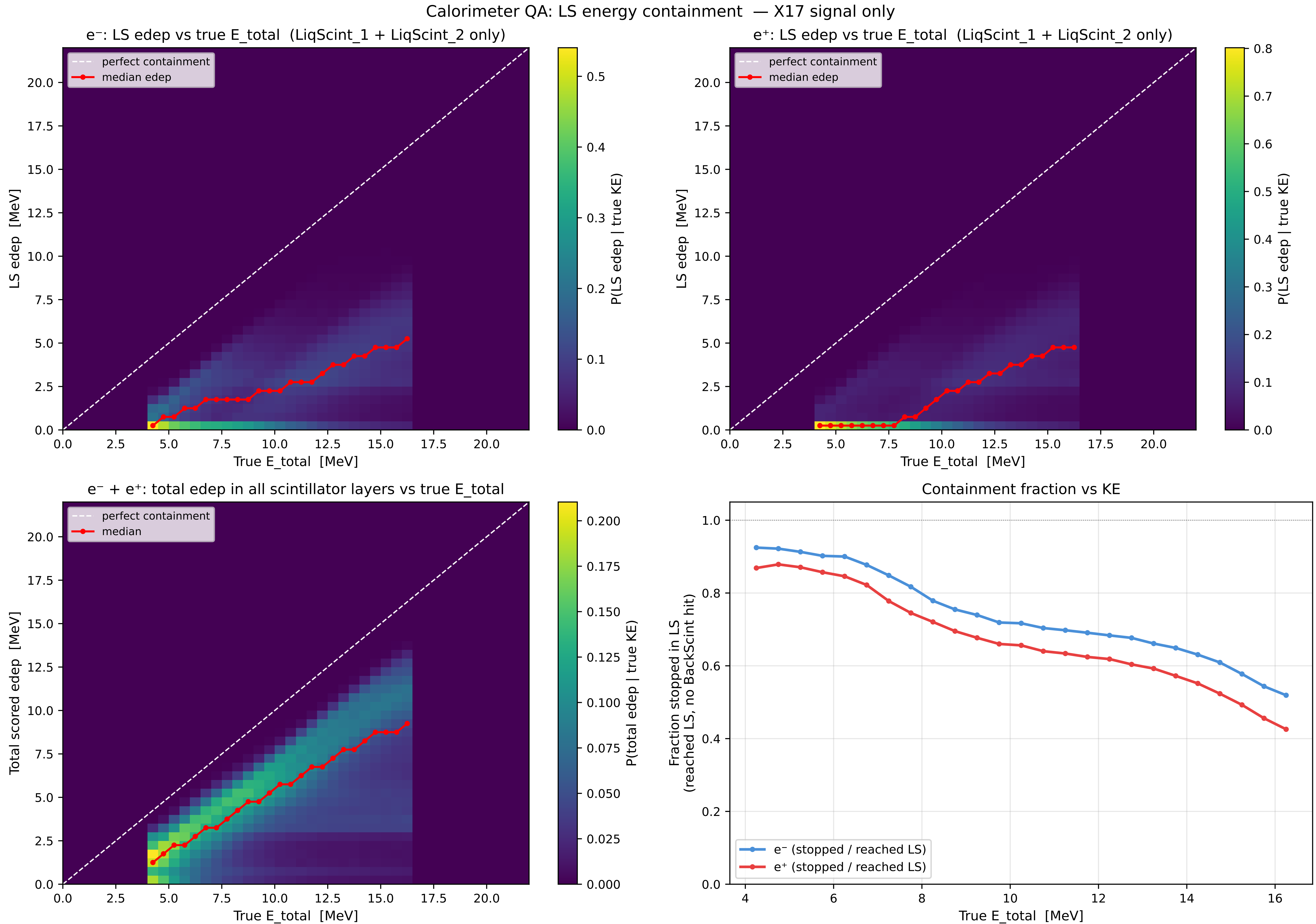


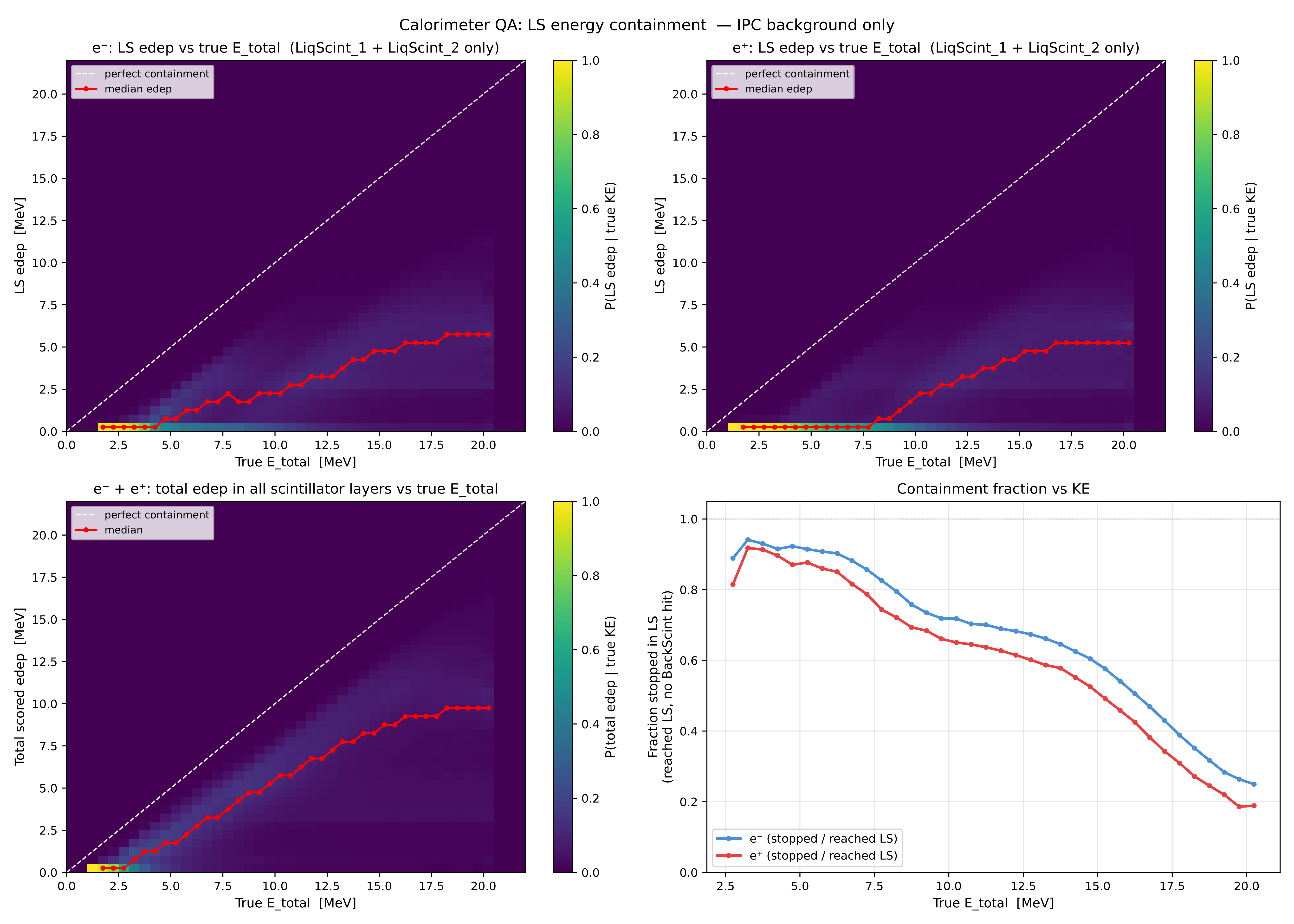
Invariant Mass (exact formula including mₑ)
Truth invariant mass
X17 signal (N=5,000,686)
IPC background (N=4,999,314)
m_X17 = 16.8 MeV
Normalised counts
True e⁺e⁻ invariant mass [MeV]
Reco mass — LS only (uncorrected energy)
exact: $M = \sqrt{2m_e^2 + 2(E_{e^-}E_{e^+} - p_{e^-}p_{e^+}\cos\theta)}$
X17 signal (N=97,275)
IPC background (N=110,100)
m_X17 = 16.8 MeV
Reco invariant mass [MeV]
Reco mass — all scintillators
(PlScint + LS + BackScint energy)
X17 signal (N=640,511)
IPC background (N=643,687)
m_X17 = 16.8 MeV
Reco invariant mass [MeV]



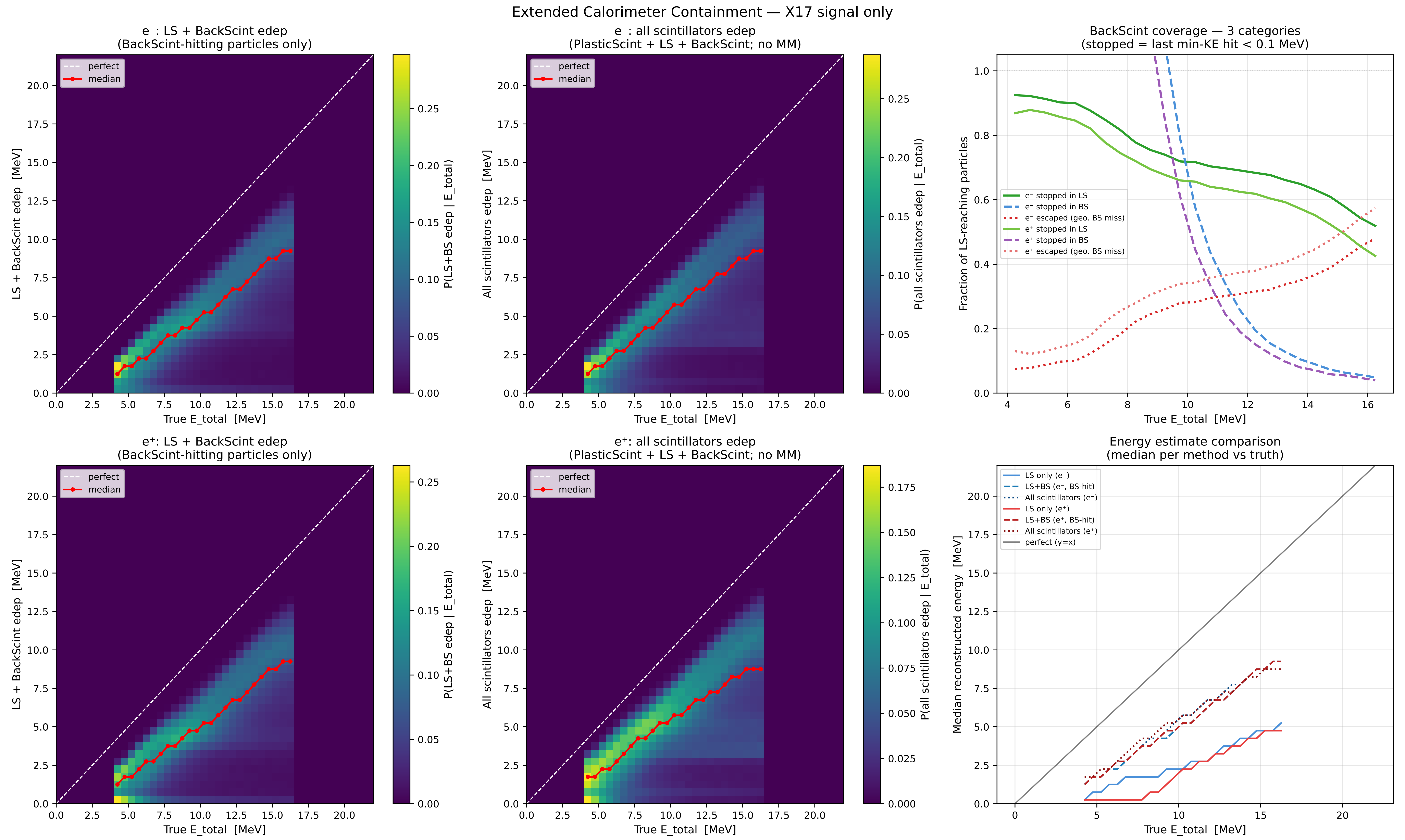
X17 signal: energy-corrected invariant mass
Correction: $E_{est} = (E_{dep} + b)/(1 - a)$ from upstream-loss fit $\Longrightarrow$ $M_{corr} \approx M_{reco}/\sqrt{(1 - a_{e^-})(1 - a_{e^+})}$ [large-E approx]
Truth
Reco — LS only
Reco — all scintillators
LS corrected (×3.20)
All scint. corrected (×1.43)
m_X17 = 16.8 MeV
Normalised counts
0.00
0.02
0.04
0.06
0.08
0
10
20
30
40
50
60
70
80
Invariant mass [MeV]


Calorimeter QA: LS energy containment — X17 + IPC combined
e⁻: LS edep vs true E_total (LiqScint_1 + LiqScint_2 only)
e⁺: LS edep vs true E_total (LiqScint_1 + LiqScint_2 only)
perfect containment
median edep
LS edep [MeV]
True E_total [MeV]
P(LS edep | true KE)
e⁻ + e⁺: total edep in all scintillator layers vs true E_total
median
Total scored edep [MeV]
P(total edep | true KE)
Containment fraction vs KE
Fraction stopped in LS
(reached LS, no BackScint hit)
e⁻ (stopped / reached LS)
e⁺ (stopped / reached LS)

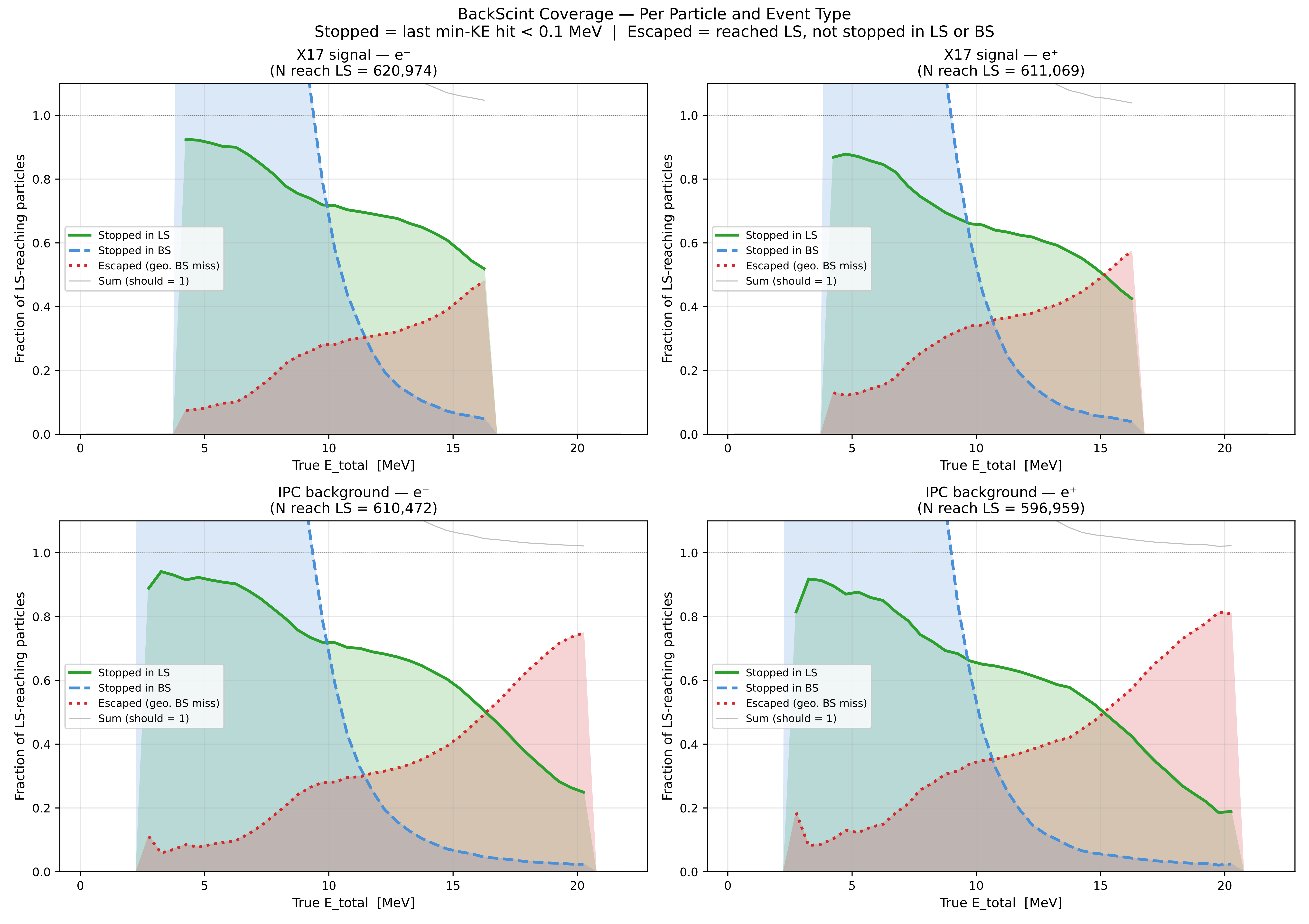

Calorimeter QA: LS energy containment — X17 signal only
e⁻: LS edep vs true E_total (LiqScint_1 + LiqScint_2 only)
e⁺: LS edep vs true E_total (LiqScint_1 + LiqScint_2 only)
perfect containment
median edep
LS edep [MeV]
True E_total [MeV]
P(LS edep | true KE)
e⁻ + e⁺: total edep in all scintillator layers vs true E_total
median
Total scored edep [MeV]
P(total edep | true KE)
Containment fraction vs KE
Fraction stopped in LS
(reached LS, no BackScint hit)
e⁻ (stopped / reached LS)
e⁺ (stopped / reached LS)



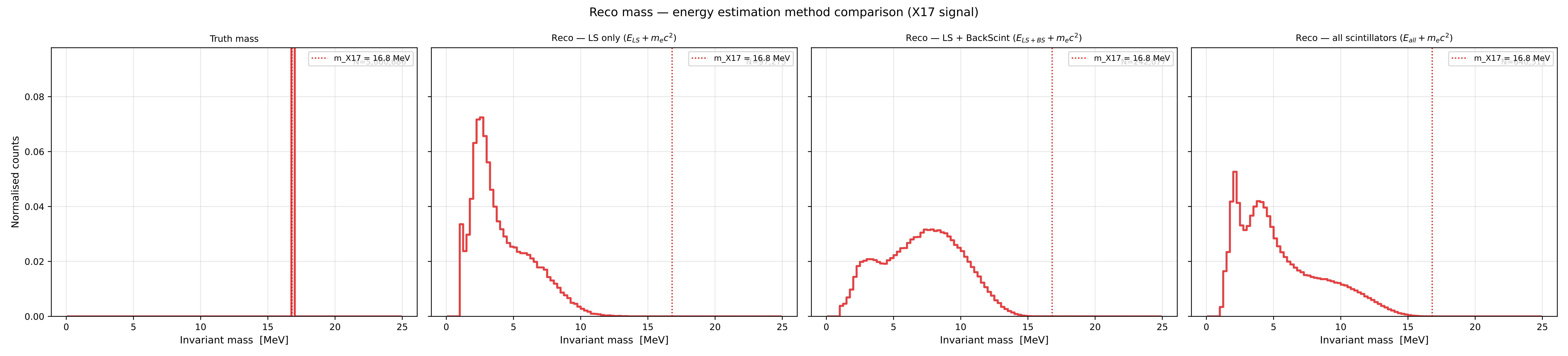
Calorimeter QA: LS energy containment — IPC background only
e⁻: LS edep vs true E_total (LiqScint_1 + LiqScint_2 only)
e⁺: LS edep vs true E_total (LiqScint_1 + LiqScint_2 only)
perfect containment
median edep
LS edep [MeV]
True E_total [MeV]
P(LS edep | true KE)
e⁻ + e⁺: total edep in all scintillator layers vs true E_total
median
Total scored edep [MeV]
P(total edep | true KE)
Containment fraction vs KE
Fraction stopped in LS
(reached LS, no BackScint hit)
e⁻ (stopped / reached LS)
e⁺ (stopped / reached LS)



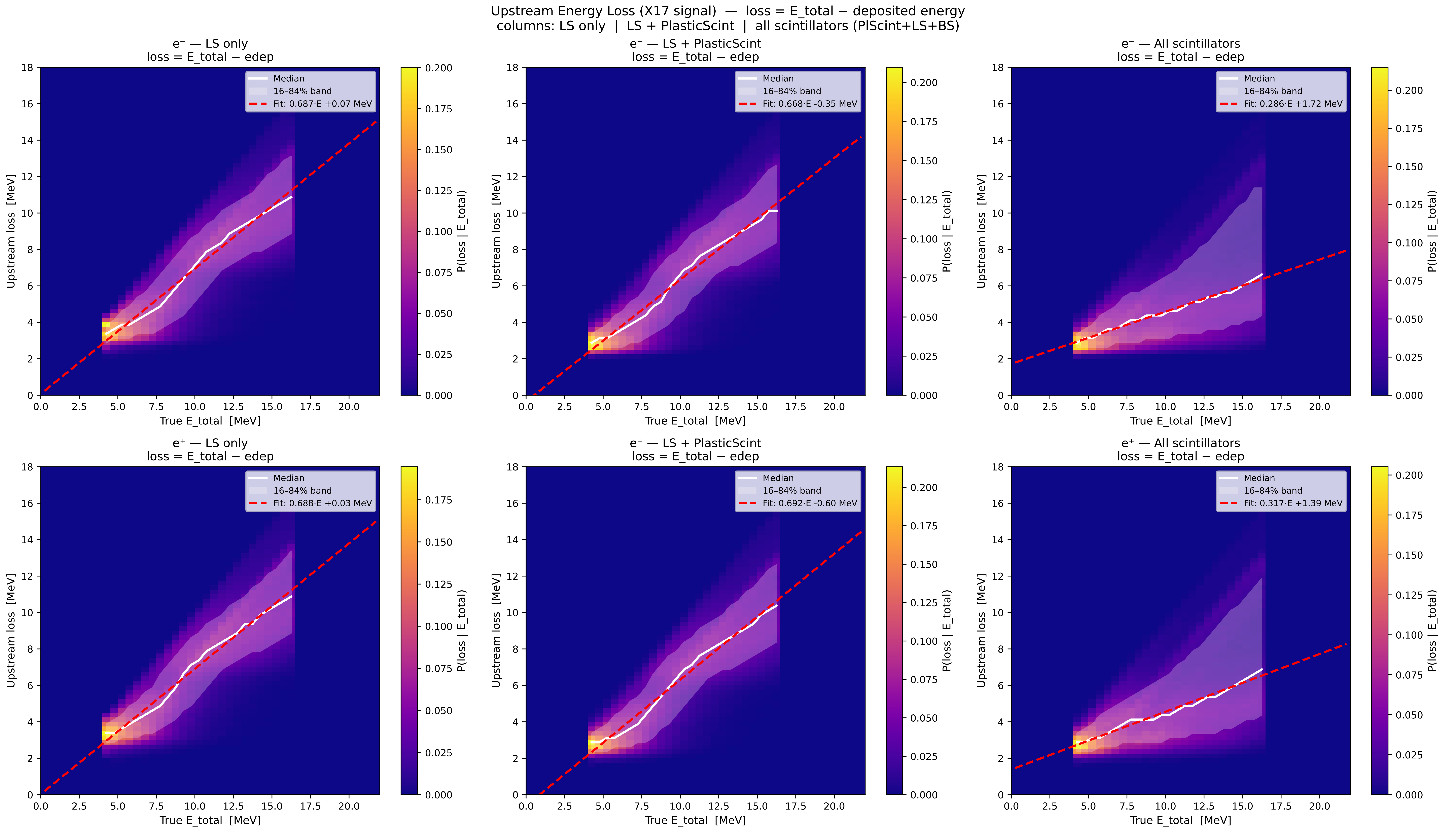
Extended Calorimeter Containment — X17 signal only
e⁻: LS + BackScint edep
(BackScint-hitting particles only)
e⁻: all scintillators edep
(PlasticScint + LS + BackScint; no MM)
BackScint coverage — 3 categories
(stopped = last min-KE hit < 0.1 MeV)
e⁺: LS + BackScint edep
(BackScint-hitting particles only)
e⁺: all scintillators edep
(PlasticScint + LS + BackScint; no MM)
Energy estimate comparison
(median per method vs truth)
perfect
median
True E_total [MeV]
LS + BackScint edep [MeV]
All scintillators edep [MeV]
P(LS+BS edep | E_total)
P(all scintillators edep | E_total)
Fraction of LS-reaching particles
e⁻ stopped in LS
e⁻ stopped in BS
e⁻ escaped (geo. BS miss)
e⁺ stopped in LS
e⁺ stopped in BS
e⁺ escaped (geo. BS miss)
Median reconstructed energy [MeV]
LS only (e⁻)
LS+BS (e⁻, BS-hit)
All scintillators (e⁻)
LS only (e⁺)
LS+BS (e⁺, BS-hit)
All scintillators (e⁺)
perfect (y=x)



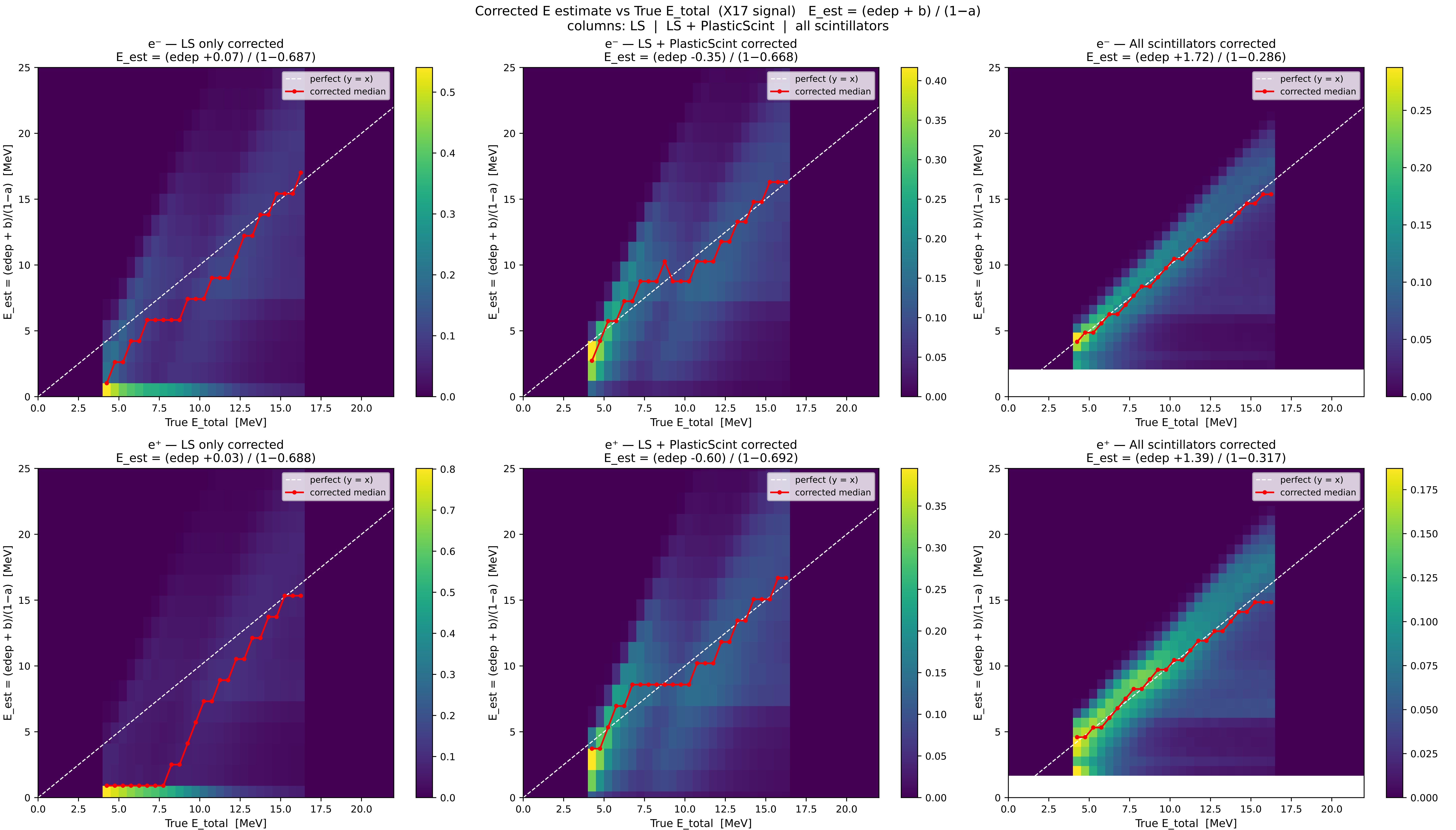
BackScint Coverage — Per Particle and Event Type
Stopped = last min-KE hit < 0.1 MeV | Escaped = reached LS, not stopped in LS or BS
X17 signal — e⁻
(N reach LS = 620,974)
X17 signal — e⁺
(N reach LS = 611,069)
IPC background — e⁻
(N reach LS = 610,472)
IPC background — e⁺
(N reach LS = 596,959)
Stopped in LS
Stopped in BS
Escaped (geo. BS miss)
Sum (should = 1)
Fraction of LS-reaching particles
True E_total [MeV]

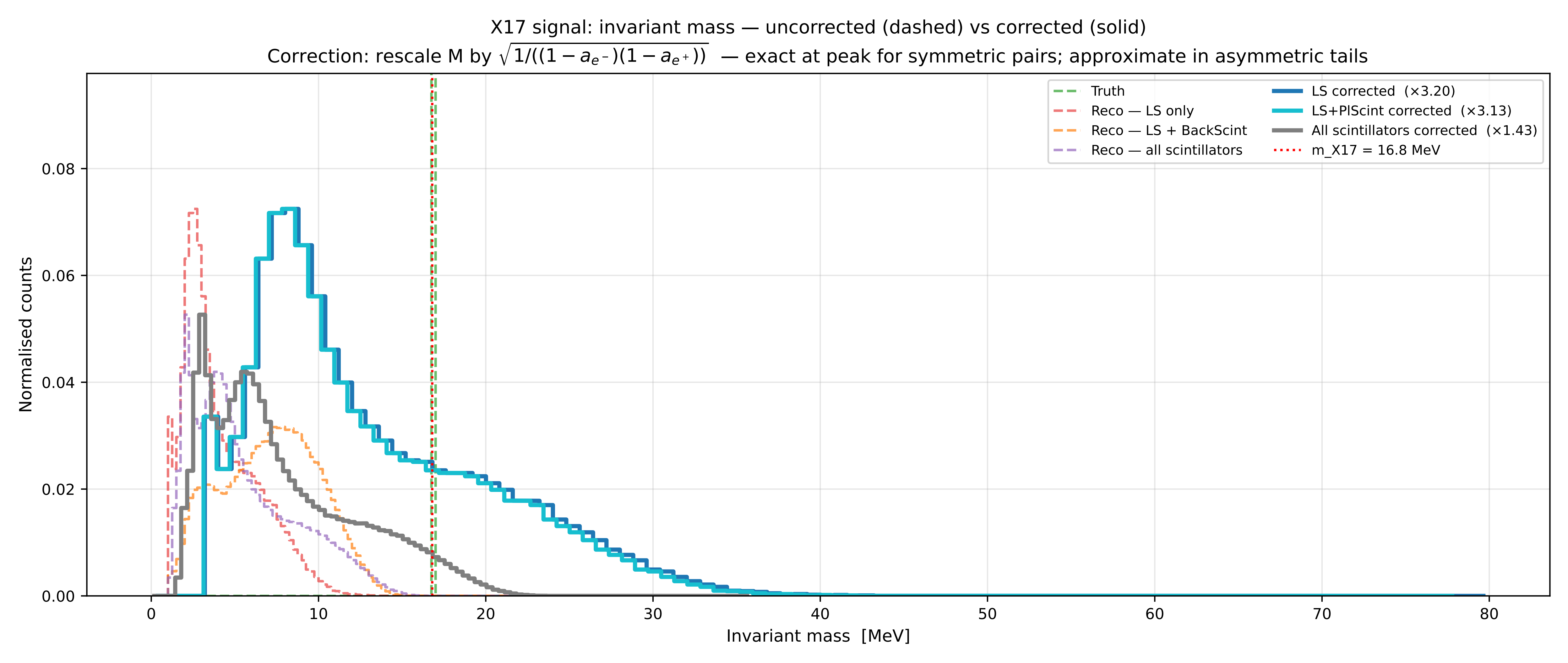


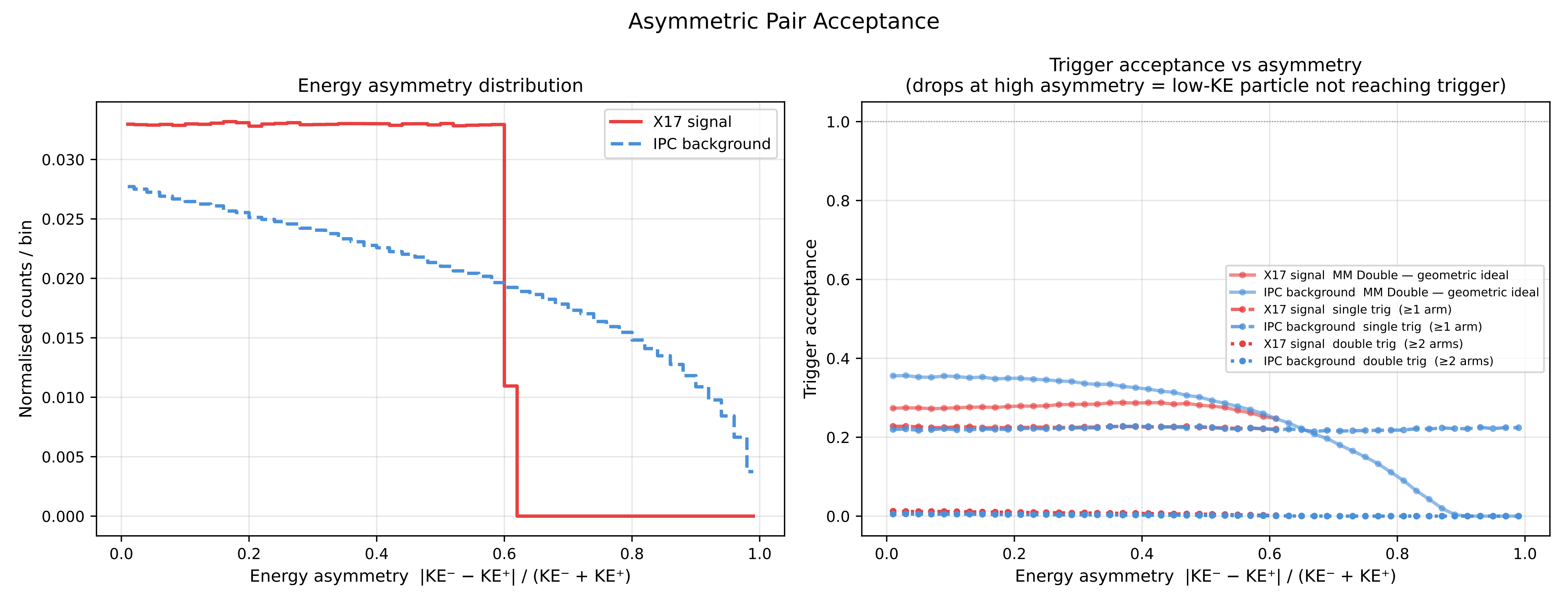
Reco mass — energy estimation method comparison (X17 signal)
Truth mass
Reco — LS only ($E_{LS} + m_e c^2$)
Reco — LS + BackScint ($E_{LS+BS} + m_e c^2$)
Reco — all scintillators ($E_{all} + m_e c^2$)
m_X17 = 16.8 MeV
Normalised counts
Invariant mass [MeV]
0.00
0.02
0.04
0.06
0.08
0
5
10
15
20
25

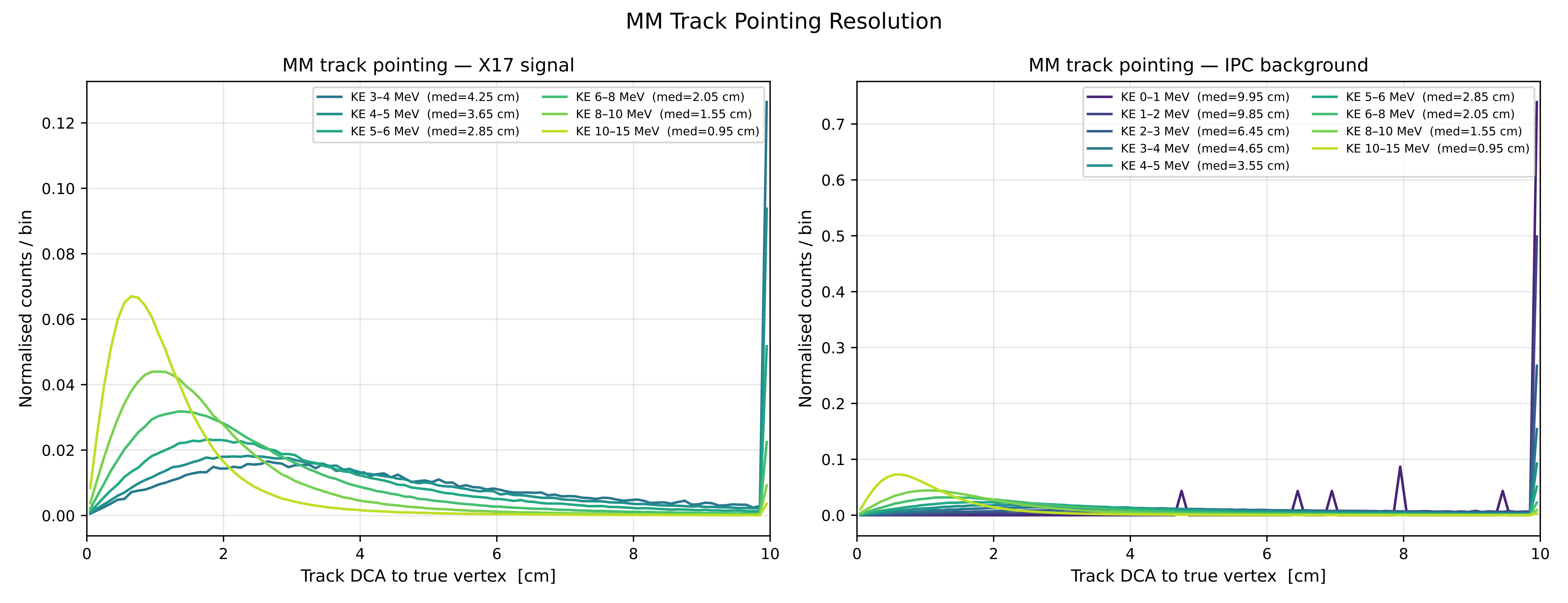


Upstream Energy Loss (X17 signal) — loss = E_total − deposited energy
columns: LS only | LS + PlasticScint | all scintillators (PlScint+LS+BS)
e⁻ — LS only
loss = E_total − edep
Median
16–84% band
Fit: 0.687·E +0.07 MeV
e⁻ — LS + PlasticScint
loss = E_total − edep
Median
16–84% band
Fit: 0.668·E -0.35 MeV
e⁻ — All scintillators
loss = E_total − edep
Median
16–84% band
Fit: 0.286·E +1.72 MeV
e⁺ — LS only
loss = E_total − edep
Median
16–84% band
Fit: 0.688·E +0.03 MeV
e⁺ — LS + PlasticScint
loss = E_total − edep
Median
16–84% band
Fit: 0.692·E -0.60 MeV
e⁺ — All scintillators
loss = E_total − edep
Median
16–84% band
Fit: 0.317·E +1.39 MeV
Upstream loss [MeV]
True E_total [MeV]
P(loss | E_total)

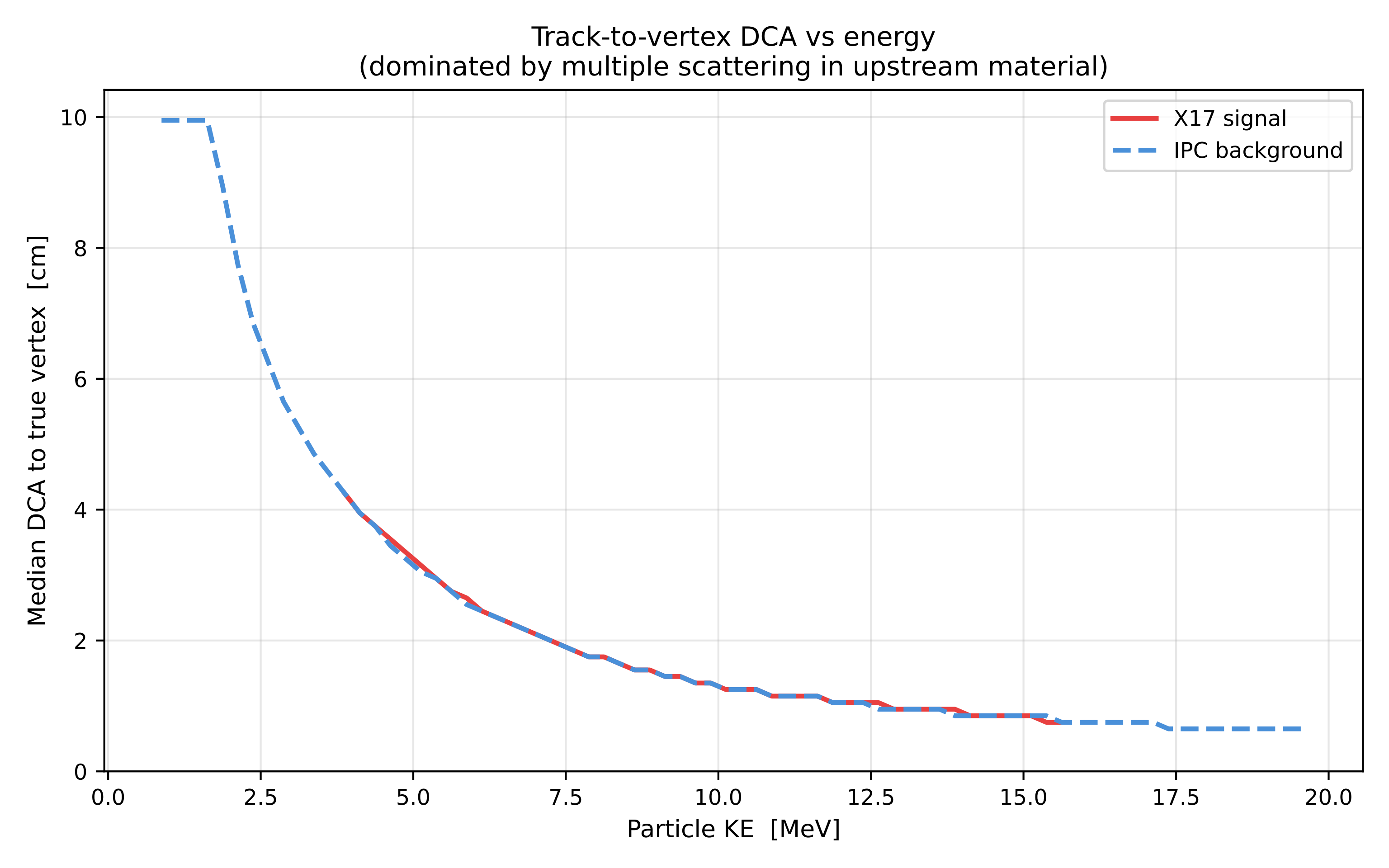

Corrected E estimate vs True E_total (X17 signal) E_est = (edep + b) / (1−a)
columns: LS | LS + PlasticScint | all scintillators
e⁻ — LS only corrected
E_est = (edep +0.07) / (1−0.687)
e⁻ — LS + PlasticScint corrected
E_est = (edep -0.35) / (1−0.668)
e⁻ — All scintillators corrected
E_est = (edep +1.72) / (1−0.286)
e⁺ — LS only corrected
E_est = (edep +0.03) / (1−0.688)
e⁺ — LS + PlasticScint corrected
E_est = (edep -0.60) / (1−0.692)
e⁺ — All scintillators corrected
E_est = (edep +1.39) / (1−0.317)
perfect (y = x)
corrected median
E_est = (edep + b)/(1−a) [MeV]
True E_total [MeV]


X17 signal: invariant mass — uncorrected (dashed) vs corrected (solid)
Correction: rescale M by $\sqrt{1/((1-a_{e^-})(1-a_{e^+}))}$ — exact at peak for symmetric pairs; approximate in asymmetric tails
Truth
Reco — LS only
Reco — LS + BackScint
Reco — all scintillators
LS corrected (×3.20)
LS+PlScint corrected (×3.13)
All scintillators corrected (×1.43)
m_X17 = 16.8 MeV
Normalised counts
0.00
0.02
0.04
0.06
0.08
0
10
20
30
40
50
60
70
80
Invariant mass [MeV]

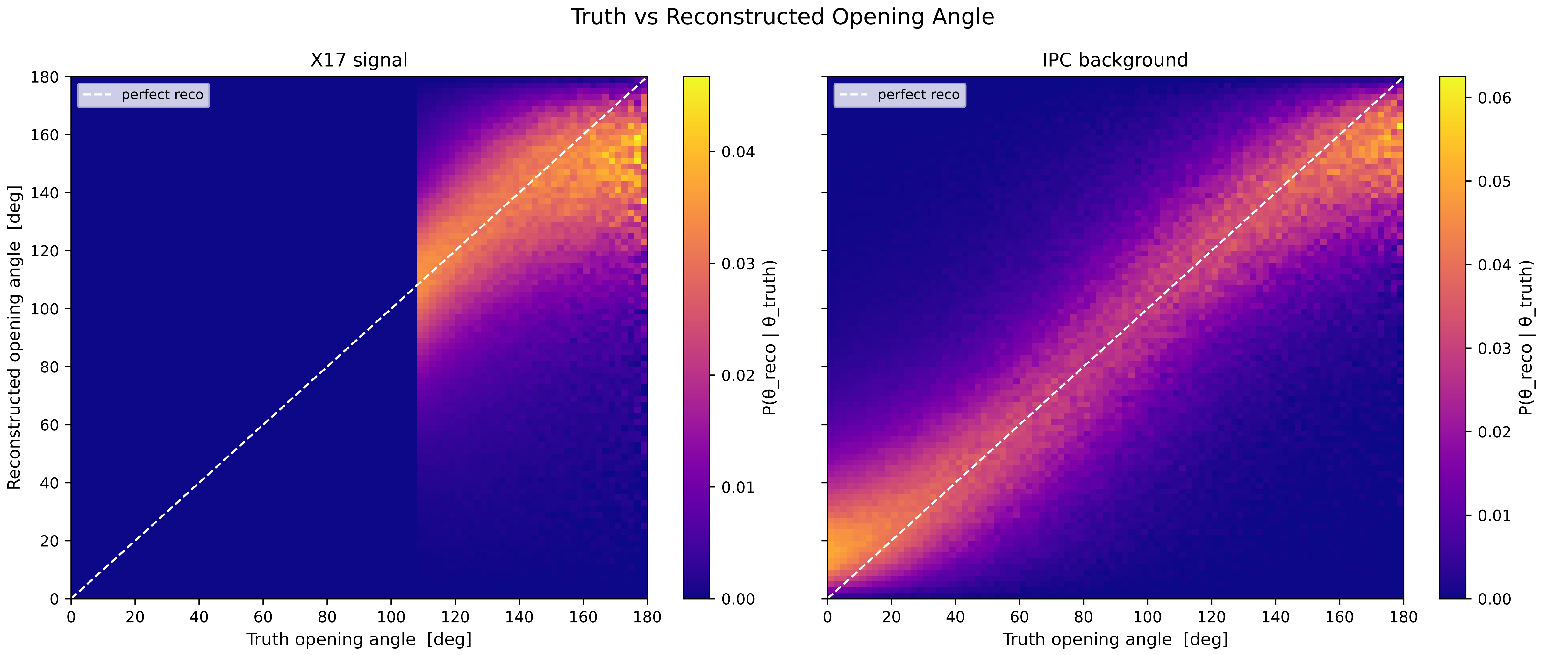

Asymmetric Pair Acceptance
Energy asymmetry distribution
X17 signal
IPC background
Normalised counts / bin
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.0
0.2
0.4
0.6
0.8
1.0
Energy asymmetry |KE⁻ − KE⁺| / (KE⁻ + KE⁺)
Trigger acceptance vs asymmetry
(drops at high asymmetry = low-KE particle not reaching trigger)
Trigger acceptance
X17 signal MM Double — geometric ideal
IPC background MM Double — geometric ideal
X17 signal single trig (≥1 arm)
IPC background single trig (≥1 arm)
X17 signal double trig (≥2 arms)
IPC background double trig (≥2 arms)
Energy asymmetry |KE⁻ − KE⁺| / (KE⁻ + KE⁺)



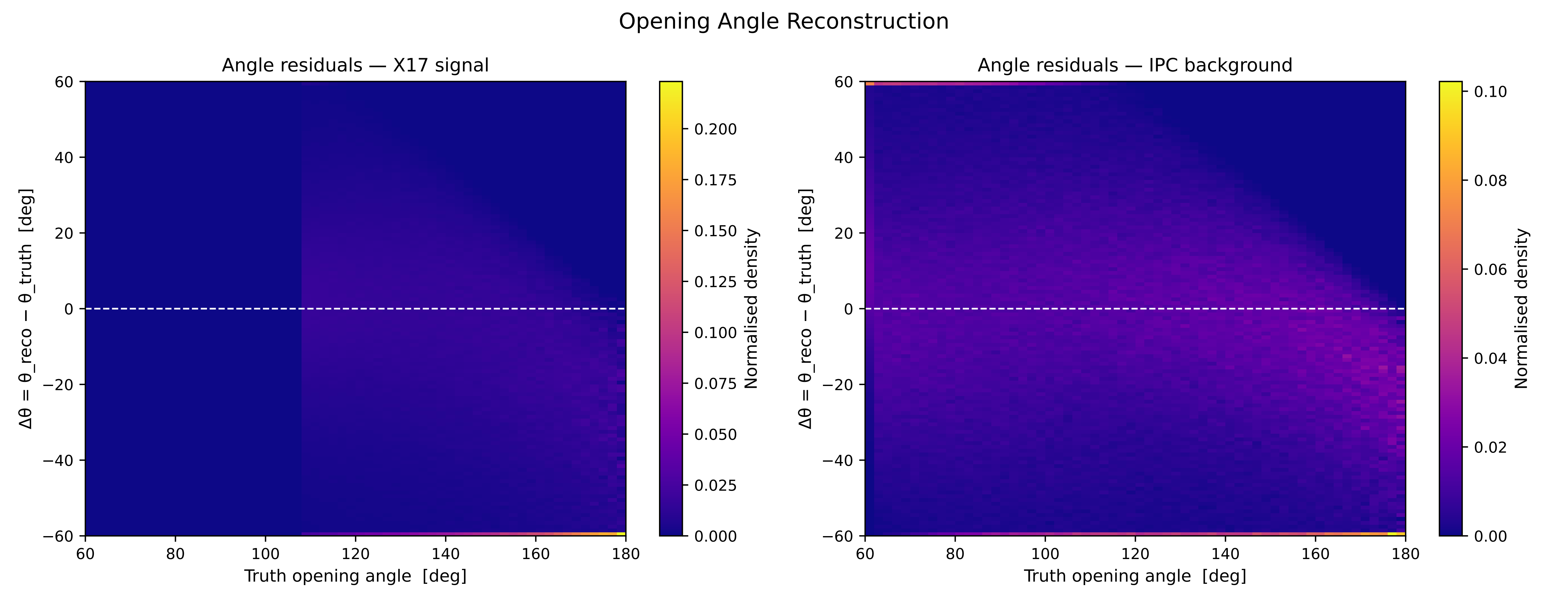
MM Track Pointing Resolution
MM track pointing — X17 signal
KE 3–4 MeV (med=4.25 cm)
KE 4–5 MeV (med=3.65 cm)
KE 5–6 MeV (med=2.85 cm)
KE 6–8 MeV (med=2.05 cm)
KE 8–10 MeV (med=1.55 cm)
KE 10–15 MeV (med=0.95 cm)
Normalised counts / bin
Track DCA to true vertex [cm]
MM track pointing — IPC background
KE 0–1 MeV (med=9.95 cm)
KE 1–2 MeV (med=9.85 cm)
KE 2–3 MeV (med=6.45 cm)
KE 3–4 MeV (med=4.65 cm)
KE 4–5 MeV (med=3.55 cm)
KE 5–6 MeV (med=2.85 cm)
KE 6–8 MeV (med=2.05 cm)
KE 8–10 MeV (med=1.55 cm)
KE 10–15 MeV (med=0.95 cm)
Normalised counts / bin
Track DCA to true vertex [cm]



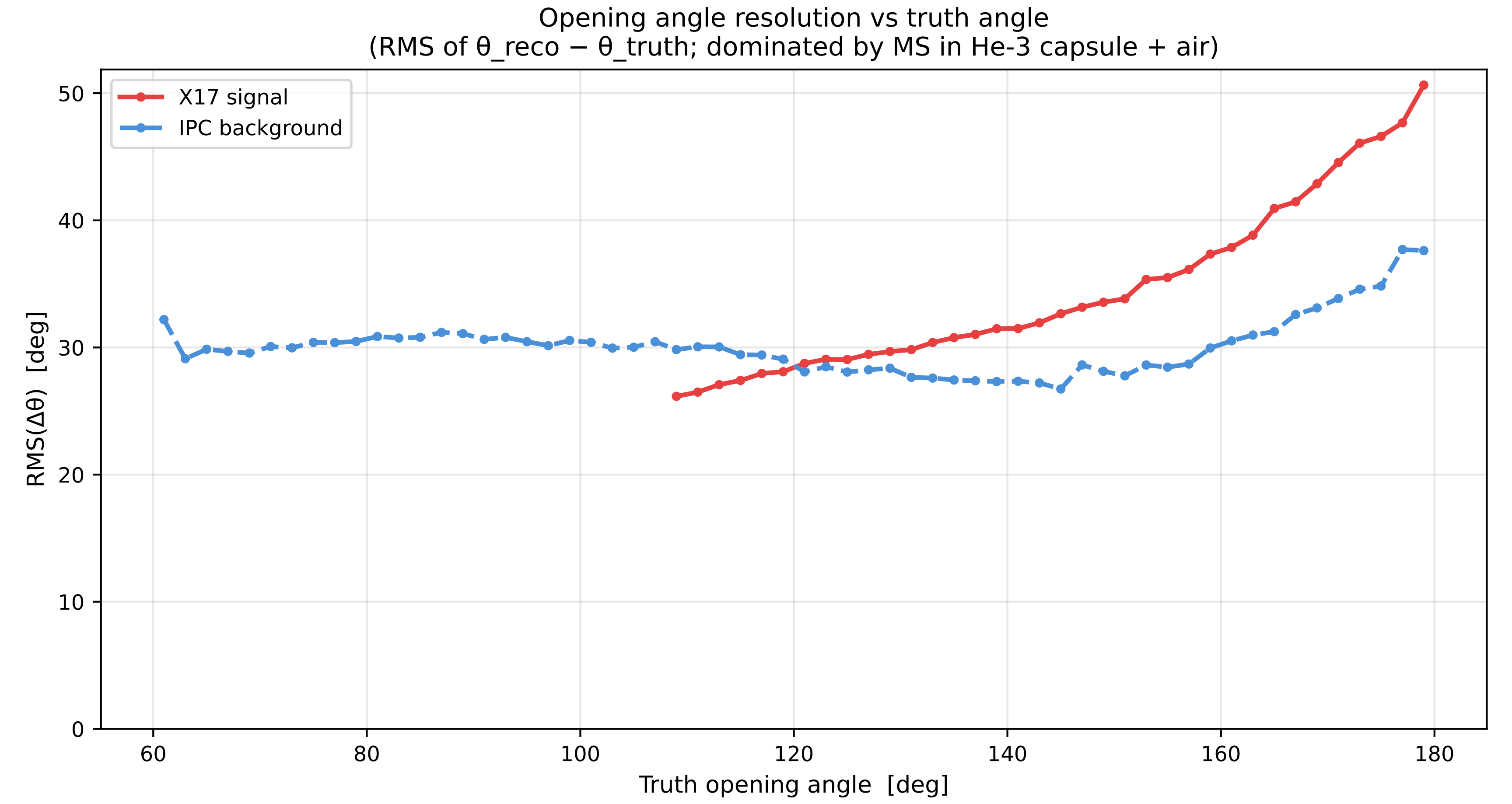
Track-to-vertex DCA vs energy
(dominated by multiple scattering in upstream material)
X17 signal
IPC background
Median DCA to true vertex [cm]
Particle KE [MeV]
0
2
4
6
8
10
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



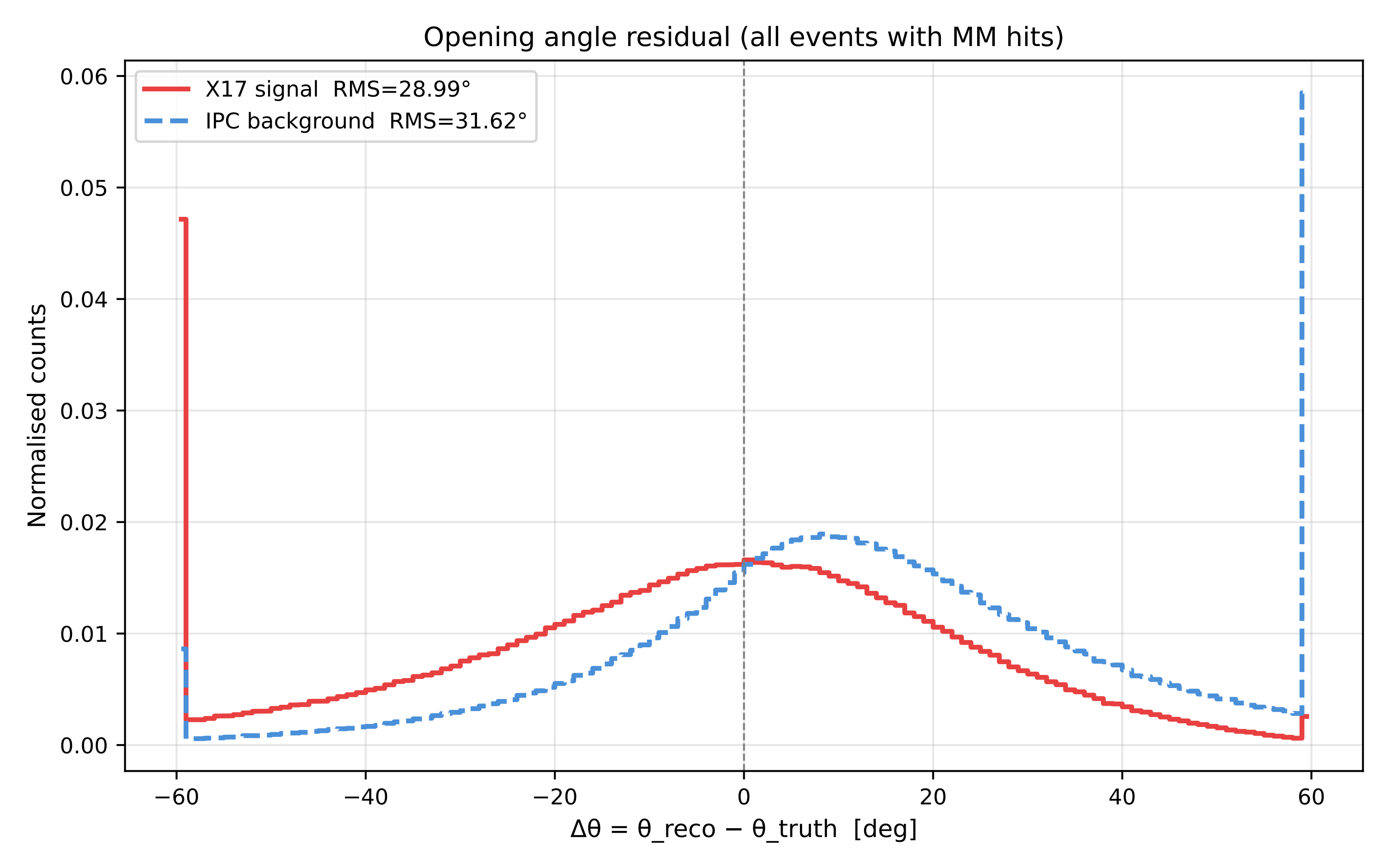
Truth vs Reconstructed Opening Angle
X17 signal
perfect reco
Reconstructed opening angle [deg]
Truth opening angle [deg]
P(θ_reco | θ_truth)
0
20
40
60
80
100
120
140
160
180
0.00
0.01
0.02
0.03
0.04
IPC background
perfect reco
Truth opening angle [deg]
P(θ_reco | θ_truth)
0.00
0.01
0.02
0.03
0.04
0.05
0.06

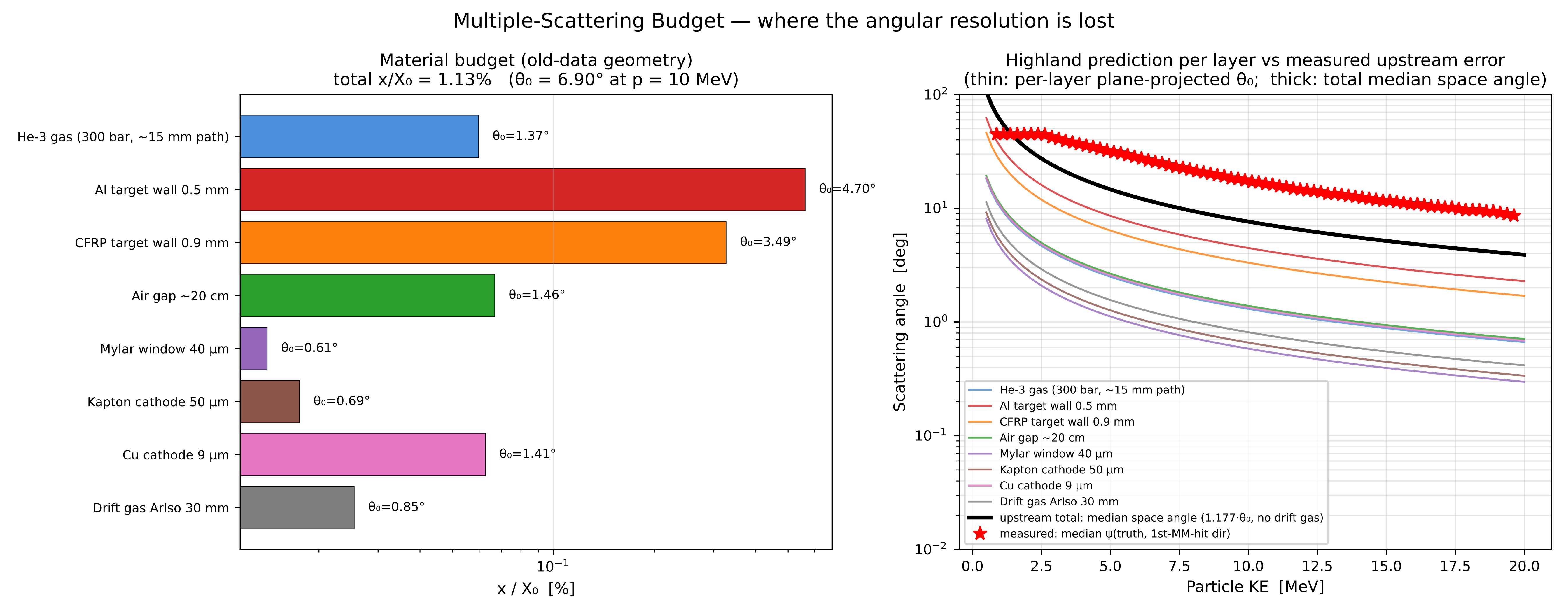


Opening Angle Reconstruction
Angle residuals — X17 signal
Angle residuals — IPC background
Δθ = θ_reco − θ_truth [deg]
Truth opening angle [deg]
Normalised density
60
40
20
0
−20
−40
−60
80
100
120
140
160
180
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
0.00
0.02
0.04
0.06
0.08
0.10

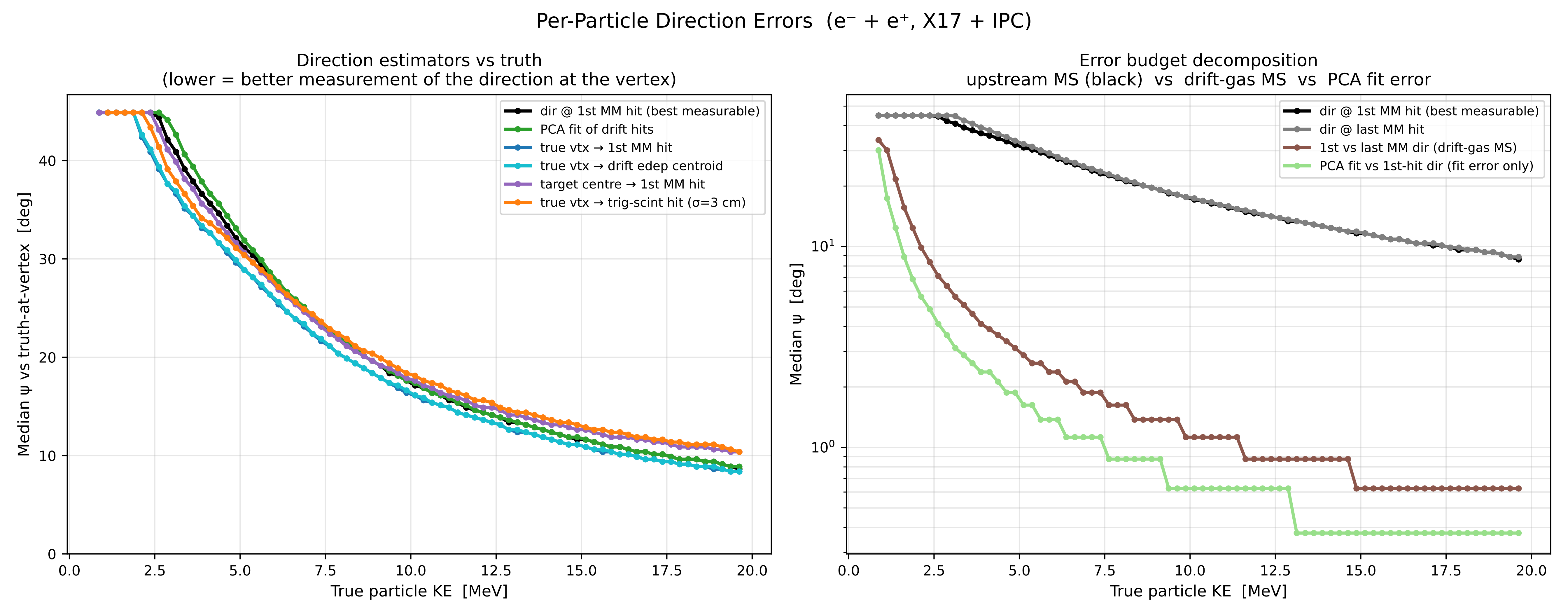


Opening angle resolution vs truth angle
(RMS of θ_reco − θ_truth; dominated by MS in He-3 capsule + air)
X17 signal
IPC background
RMS(Δθ) [deg]
Truth opening angle [deg]
0
10
20
30
40
50
60
80
100
120
140
160
180

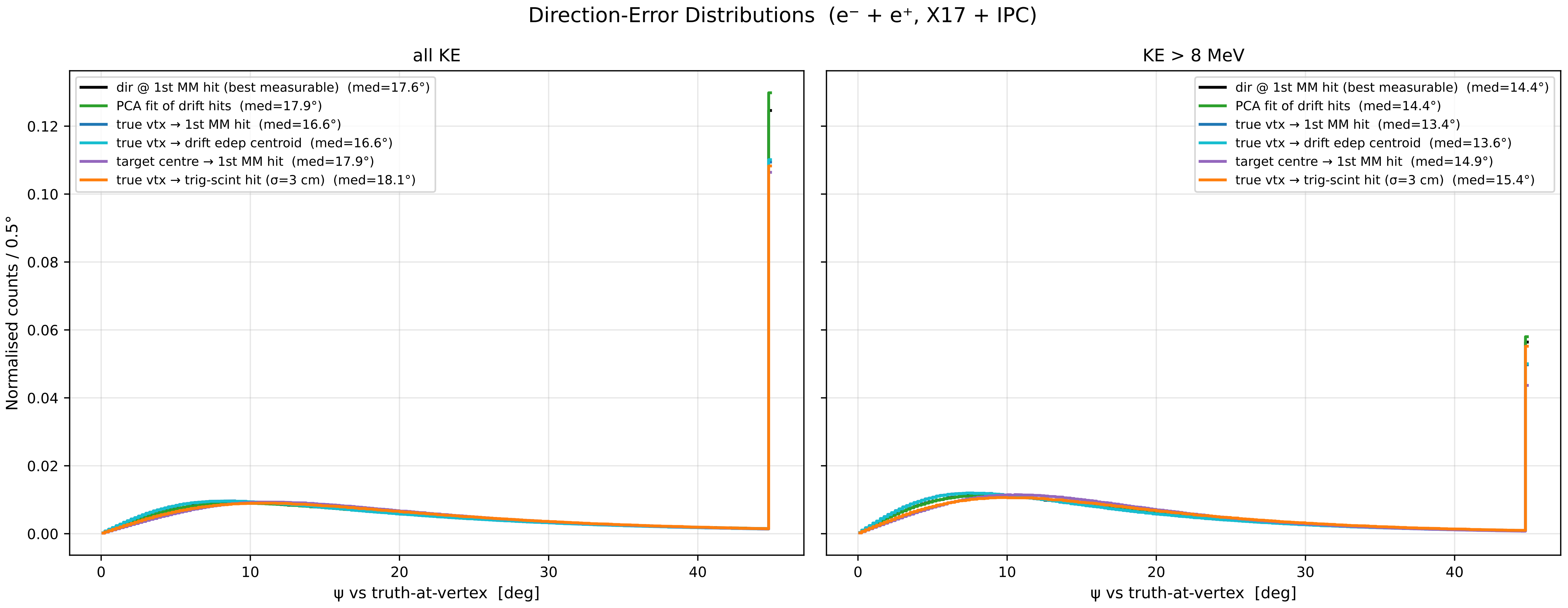

Opening angle residual (all events with MM hits)
X17 signal RMS=28.99°
IPC background RMS=31.62°
Normalised counts
0.06
0.05
0.04
0.03
0.02
0.01
0.00
−60
−40
−20
0
20
40
60
Δθ = θ_reco − θ_truth [deg]



Multiple-Scattering Budget — where the angular resolution is lost
Material budget (old-data geometry)
total x/X₀ = 1.13% (θ₀ = 6.90° at p = 10 MeV)
He-3 gas (300 bar, ~15 mm path)
Al target wall 0.5 mm
CFRP target wall 0.9 mm
Air gap ~20 cm
Mylar window 40 µm
Kapton cathode 50 µm
Cu cathode 9 µm
Drift gas ArIso 30 mm
θ₀=1.37°
θ₀=4.70°
θ₀=3.49°
θ₀=1.46°
θ₀=0.61°
θ₀=0.69°
θ₀=1.41°
θ₀=0.85°
10⁻¹
x / X₀ [%]
Highland prediction per layer vs measured upstream error
(thin: per-layer plane-projected θ₀; thick: total median space angle)
Scattering angle [deg]
Particle KE [MeV]
upstream total: median space angle (1.177·θ₀, no drift gas)
measured: median ψ(truth, 1st-MM-hit dir)

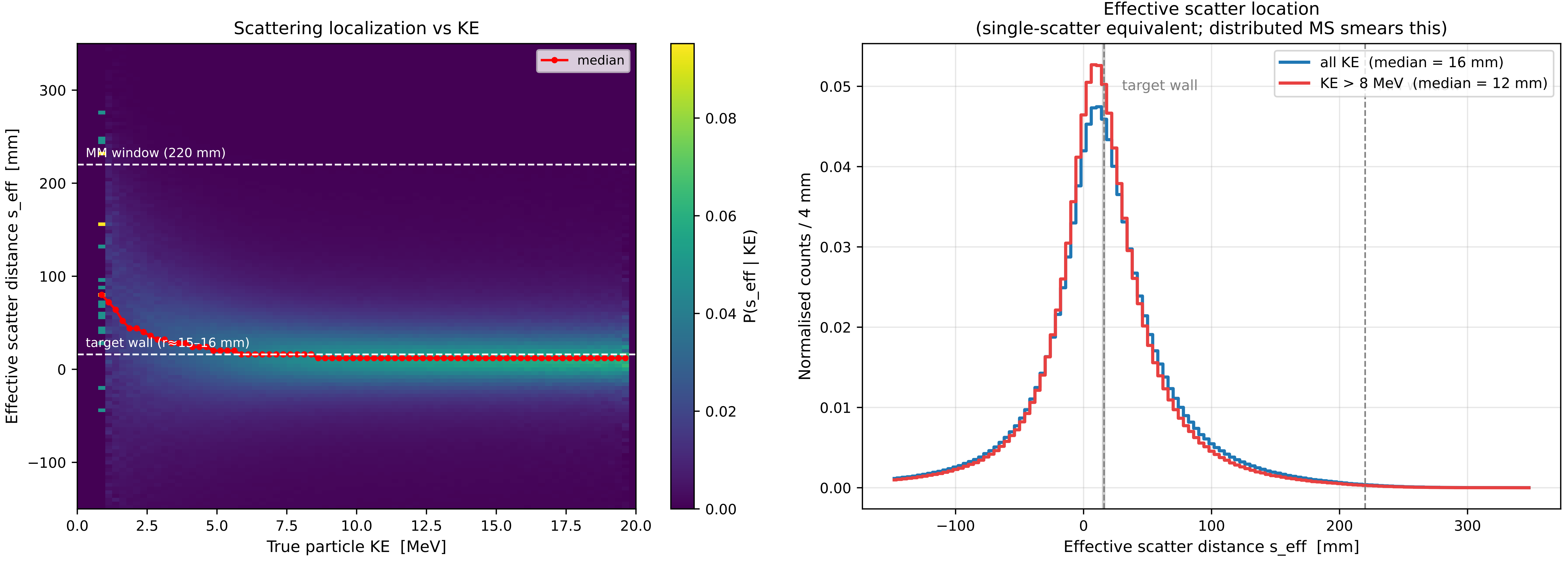


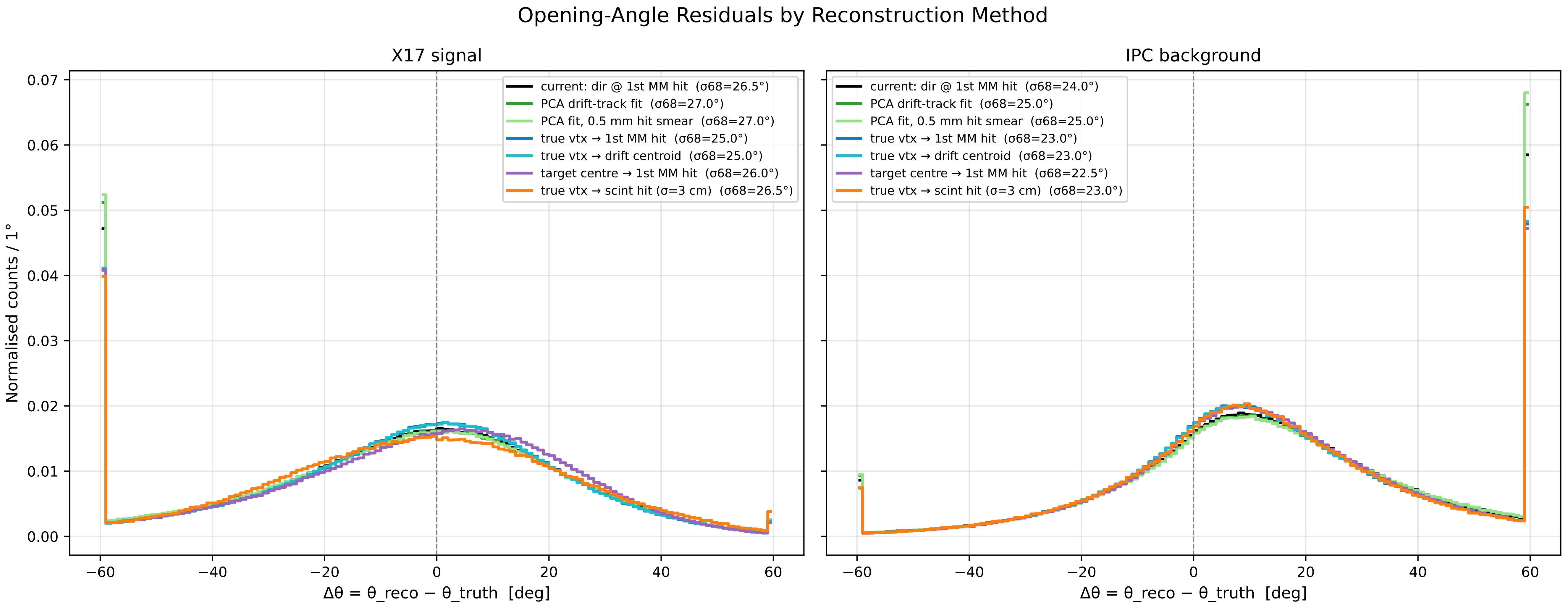
Per-Particle Direction Errors (e⁻ + e⁺, X17 + IPC)
Direction estimators vs truth
(lower = better measurement of the direction at the vertex)
dir @ 1st MM hit (best measurable)
PCA fit of drift hits
true vtx → 1st MM hit
true vtx → drift edep centroid
target centre → 1st MM hit
true vtx → trig-scint hit (σ=3 cm)
Median ψ vs truth-at-vertex [deg]
True particle KE [MeV]
Error budget decomposition
upstream MS (black) vs drift-gas MS vs PCA fit error
dir @ 1st MM hit (best measurable)
dir @ last MM hit
1st vs last MM dir (drift-gas MS)
PCA fit vs 1st-hit dir (fit error only)
Median ψ [deg]
True particle KE [MeV]



Direction-Error Distributions (e⁻ + e⁺, X17 + IPC)
all KE
KE > 8 MeV
dir @ 1st MM hit (best measurable) (med=17.6°)
PCA fit of drift hits (med=17.9°)
true vtx → 1st MM hit (med=16.6°)
true vtx → drift edep centroid (med=16.6°)
target centre → 1st MM hit (med=17.9°)
true vtx → trig-scint hit (σ=3 cm) (med=18.1°)
dir @ 1st MM hit (best measurable) (med=14.4°)
PCA fit of drift hits (med=14.4°)
true vtx → 1st MM hit (med=13.4°)
true vtx → drift edep centroid (med=13.6°)
target centre → 1st MM hit (med=14.9°)
true vtx → trig-scint hit (σ=3 cm) (med=15.4°)
Normalised counts / 0.5°
ψ vs truth-at-vertex [deg]

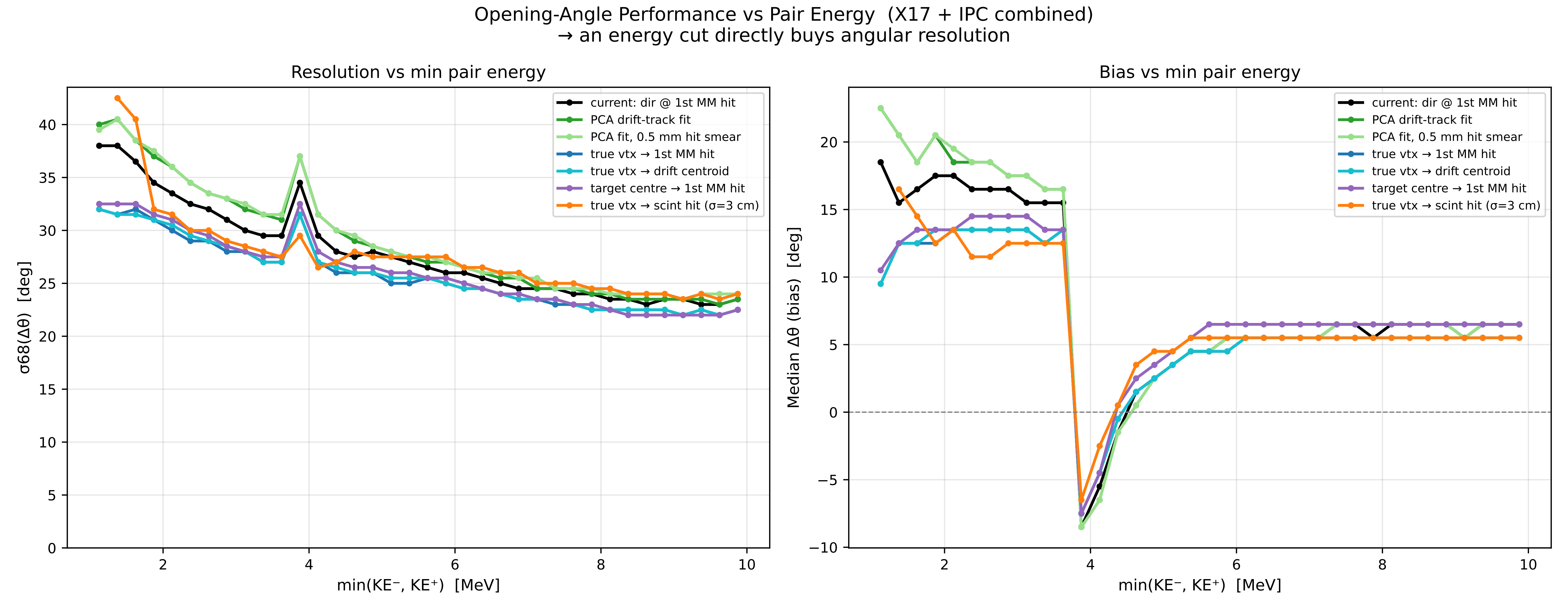
Scattering Localization (s_eff = t − δ/tanψ; requires ψ > 1.5°)



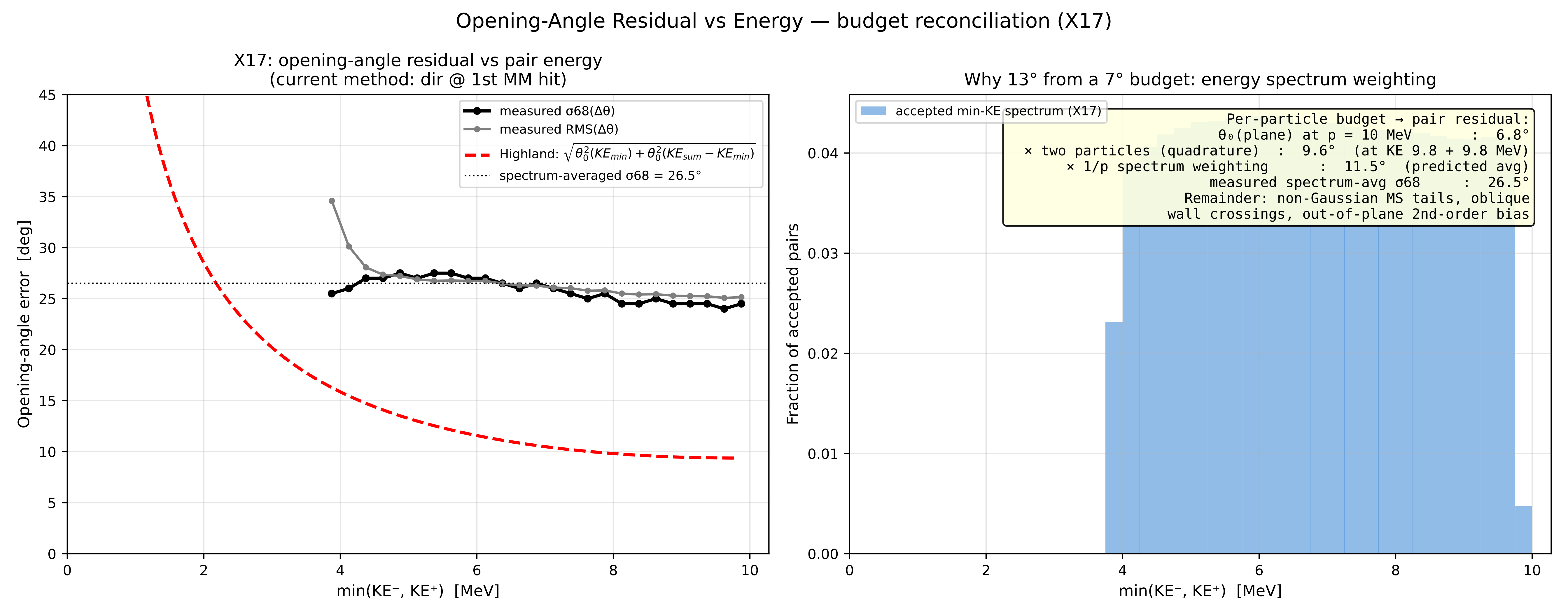


Opening-Angle Residuals by Reconstruction Method
X17 signal
IPC background
current: dir @ 1st MM hit (σ68=26.5°)
PCA drift-track fit (σ68=27.0°)
PCA fit, 0.5 mm hit smear (σ68=27.0°)
true vtx → 1st MM hit (σ68=25.0°)
true vtx → drift centroid (σ68=25.0°)
target centre → 1st MM hit (σ68=26.0°)
true vtx → scint hit (σ=3 cm) (σ68=26.5°)
current: dir @ 1st MM hit (σ68=24.0°)
PCA drift-track fit (σ68=25.0°)
PCA fit, 0.5 mm hit smear (σ68=25.0°)
true vtx → 1st MM hit (σ68=23.0°)
true vtx → drift centroid (σ68=23.0°)
target centre → 1st MM hit (σ68=22.5°)
true vtx → scint hit (σ=3 cm) (σ68=23.0°)
Normalised counts / 1°
Δθ = θ_reco − θ_truth [deg]
Δθ = θ_reco − θ_truth [deg]

Opening-Angle Resolution vs Truth Angle by Method
X17 signal
IPC background
current: dir @ 1st MM hit
PCA drift-track fit
PCA fit, 0.5 mm hit smear
true vtx → 1st MM hit
true vtx → drift centroid
target centre → 1st MM hit
true vtx → scint hit (σ=3 cm)
RMS(Δθ) [deg]
Truth opening angle [deg]



Opening-Angle Performance vs Pair Energy (X17 + IPC combined)
→ an energy cut directly buys angular resolution
Resolution vs min pair energy
σ68(Δθ) [deg]
min(KE⁻, KE⁺) [MeV]
Bias vs min pair energy
Median Δθ (bias) [deg]
min(KE⁻, KE⁺) [MeV]
current: dir @ 1st MM hit
PCA drift-track fit
PCA fit, 0.5 mm hit smear
true vtx → 1st MM hit
true vtx → drift centroid
target centre → 1st MM hit
true vtx → scint hit (σ=3 cm)



Opening-Angle Residual vs Energy — budget reconciliation (X17)
X17: opening-angle residual vs pair energy
(current method: dir @ 1st MM hit)
measured σ68(Δθ)
measured RMS(Δθ)
Highland: $\sqrt{\theta_0^2(KE_{min}) + \theta_0^2(KE_{sum} - KE_{min})}$
spectrum-averaged σ68 = 26.5°
Opening-angle error [deg]
min(KE⁻, KE⁺) [MeV]
Why 13° from a 7° budget: energy spectrum weighting
accepted min-KE spectrum (X17)
Per-particle budget → pair residual:
θ₀(plane) at p = 10 MeV : 6.8°
× two particles (quadrature) : 9.6° (at KE 9.8 + 9.8 MeV)
× 1/p spectrum weighting : 11.5° (predicted avg)
measured spectrum-avg σ68 : 26.5°
Remainder: non-Gaussian MS tails, oblique
wall crossings, out-of-plane 2nd-order bias
Fraction of accepted pairs
min(KE⁻, KE⁺) [MeV]